

Think Global, Invest Responsible: Why the Private Equity Industry Goes Green

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Abstract The growth of socially responsible investment (SRI) on public financial markets has drawn considerable academic attention over the last decade. Discarding from the previous literature, this article sets up to analyze the Private Equity channel, which is shown to have the potentiality to foster sustainable practices in unlisted companies. The fast integration of the environmental, social and governance issues by mainstream Private Equity investors is unveiled and appears to have benefited from the maturation of SRI on public financial markets and the impetus of large conventional actors. Hypotheses on the characteristics and drivers of this movement are proposed and tested on a unique database covering the French Private Equity industry in 2011. Empirical findings support that Private Equity socially responsible investing is characterized by investor engagement and strategically driven by a need for new value creation sources, increased risk management and differentiation. In particular, results show that independent funds, which need to attract investors, are more likely than captive funds to

develop socially responsible practices. Evolution of the movement and future research paths are proposed.

Keywords Corporate finance · Corporate Social Responsibility · France · Multi-factorial analysis · Private Equity · Engagement · Socially responsible investment · Strategy

Abbreviations

CSR	Corporate Social Responsibility
ESG	Environmental, social, and governance
LBO	Leveraged buyouts
SRI	Socially responsible investment
UNPRI	United Nations Principles for Responsible Investing

Introduction

Socially Responsible Investment (SRI) is an investment process that integrates social, environmental, and ethical considerations into investment decision-making (Renneboog et al. 2008). As such, it differs from conventional investments in a twofold way. First, socially responsible investors apply a set of investment screens to select or exclude assets based on non-financial criteria, in addition to financial criteria. Second, those investors often engage in shareholder activism to foster Corporate Social Responsibility (CSR) strategies in the firms they own. Over the last decade, the SRI market has kept expanding, reaching in 2010 about 3,070 billion USD in the United States, representing 12.2 % of assets under management (Social Investment Forum 2010) and 3,800 billion EUR in Europe in 2010 (Eurosif 2010). SRI markets are also expanding in Canada (EIA 2005), Australia (SIO 2005), and Asia (ASRIA 2011).

Consequently, SRI structure, performance, and evolution triggered research in management, economics, and

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finance fields. However, most academic work focuses on SRI provided on public financial markets. The potential impact of the Private Equity channel on a firm's non-financial policies and performance has received less attention (Scholtens 2006; Cumming and Johan 2007), partly because it is still at its early steps. Discarding from the previous literature, this article unveils responsible investment practices among Private Equity investors and their specificities compared to both standard Private Equity and public markets' SRI. Light is shed on their surprisingly fast integration of environment, social, and governance (ESG) criteria and drivers of this movement are empirically analyzed in the French context.

Here, the term 'Private Equity' industry refers to specialized investment firms whose business is to invest in unlisted companies, thus encompassing both venture capital and buyouts. On a theoretical level, Private Equity has been identified as highly efficient at maximizing shareholders' value by reducing agency costs and providing strong incentive to management (e.g., Jensen 1986, 1989; Kaplan and Strömberg 2009). As such, combining extra-financial and financial consideration through CSR does not appear in direct line with Private Equity investors' practices as it would amount to spend cash flows to provide public good. In fact, on an empirical level, the phenomenal growth curve of the Private Equity industry in Europe and the United States over the 2000–2007 period drew media and regulatory concerns about its effect on business and society. In particular, buyouts have been pointed out for their potentially negative social impact (Financial Services Authority 2006; Bocquet Report 2007).

Hereby, the question of the characterization of a SRI movement in Private Equity is by far not trivial and the analysis of its drivers is challenging both empirically and theoretically. The pioneering work of Cumming and Johan (2007), first to consider the direct intersection between SRI and Private Equity, analyzes the factors that influence institutional investors to allocate capital to socially responsible Private Equity investments. The authors forecast an increasing demand by institutional investors to invest responsibly and call for further research on the factors that give rise to fund managers offering such investment alternatives to their investors.

This article hence sets up to analyze the responsible investment movement in Private Equity. First, it establishes that the industry benefited from the maturation of SRI on public financial markets. In particular, the Private Equity approach of SRI appears to be a mainstream approach initiated by large conventional players. Hypotheses on the characteristics and drivers of the responsible investment movement in Private Equity are econometrically tested with multivariate analysis on a unique database covering the French Private Equity industry, third world largest after the United States and

the United Kingdom. France has also been identified as a proactive SRI market (Arjaliès 2010). Main findings are that SRI in Private Equity is characterized by investor engagement and strategically driven by a need for new value creation sources, increased risk management, and differentiation. In particular, results show that independent funds, which need to attract investors, are more likely than captive funds to integrate ESG issues in their investing practices.

The remainder of the article is organized as follows. Section "[Private Equity and responsible investment: where do we stand?](#)" defines and provides historical insights on SRI and Private Equity to establish where this industry stands in terms of socially responsible practices. Section "[Testable hypotheses on characteristics and drivers of the responsible investment movement in Private Equity](#)" builds testable hypotheses. Section "[Data and method](#)" presents data. Section "[Multivariate empirical analysis](#)" displays results. Main findings are discussed in section "[Discussion](#)" as well as the potential evolution of the socially responsible Private Equity movement in light of the upcoming regulatory context. Section "[Conclusion](#)" concludes on potential further research paths.

Private Equity and Responsible Investment: Where Do We Stand?

As a starting point, this section briefly reviews the parallel development of the SRI concept, on the one hand, and of the Private Equity industry, on the other hand, hence contextualizing the integration of responsible investment in Private Equity.

The SRI Concept, from Margin to Mainstream

SRI involves two main approaches. The first consists in using screens to either avoid (negative screening) or seek out (positive screening) specific investments when building a portfolio. Screens can be ethical, normative (based on social or environmental international norms), sectoral, or best-in-class (selecting most CSR proactive firms, whatever their sector). The second main SRI approach is engagement or shareholder activism: shareholders' voting rights are used to directly foster CSR in portfolio companies. Both approaches share the integration of a consideration of ESG issues in investment practices. Eccles and Viviers (2011) add to this definition that this integration be done "with the primary purpose of delivering higher-risk-adjusted financial returns."¹

¹ Such has not always been the case, as illustrated in Renneboog et al.'s (2008) review of the SRI maturation from the seventeenth century Quakers ideology to nowadays.

Over the 2000s, SRI shifted from a marginal niche market to a mainstream practice, a phenomenon called ‘SRI mainstreaming’ and based on the progressive penetration of SRI (extra-financial) criteria into conventional investment funds focused on financial criteria only (Sparkes and Cowton 2004; Bourghelle and Hager 2009). Dunfee (2003) concludes that SRI has the potential to become a mainstream phenomenon practiced by ordinary investors.

Louche and Lydenberg (2006) precise that this movement is likelier in the European than in the United States’ financial landscape. Indeed, Arjaliès (2010) shows that this potential has been realized by assets managers in the French market as a result of a deliberative and organized social movement, which aimed at changing the institutional logics of the asset management field. The French legislative context also mattered for the development of long-term investing and SRI in France and potentially the emergence of SRI mainstreaming: creation of a Pension trust Fund (Fonds de Reserve des Retraites) in 2001 with a specific SRI policy, Laws on Employee Savings in 2001 and 2003 promoting long-term investing, New Economic Regulation Act in 2001 and Grenelle 2 Act in 2011 introducing compulsory responsibility reporting for listed and non-listed firms. France, therefore, constitutes a very interesting market to investigate SRI in other types of assets.

The Surge and Crisis of the Private Equity Industry

Parallel to the development of the SRI market but on a whole different scale, Private Equity witnessed a striking growth curve over 2000s, until the financial crisis. To grasp the economic role of those investors and how responsible investment can fit in their practices, let us first describe their business.

Private Equity holds a key role in our economies because it finances innovation and unlisted companies, most of them being small- and medium-sized enterprises. It acts as a financial intermediary between investors (the ‘Limited Partners’) and companies. A limited partnership links the Private Equity firm, who acts as ‘General Partners’ and manages the fund, to the Limited Partners who provide the capital. Limited Partners neither manage the funds they invest in nor intervene at the investee company level, yet they regularly assess the quality of the investments made by General Partners. Typically, Limited Partners are not only institutional investors (banks, insurance companies, and pension funds) but also count family offices, individuals, corporations, and Government agencies.

Private Equity funds exist for a fixed period (usually 10 years) over which investment cycles occur: General Partners acquire companies (acquisition stage), hold them in portfolio for 4–7 years (holding stage), and sell them (exit stage) to redistribute capital and dividends to the

Limited Partners. Private Equity investors are thus significant or majority shareholders of companies for middle- to long-term horizon.²

Depending on the growth stage of the company and the complexity of the deal (Morrell and Clark 2010), the industry can be broken down to different segments. Venture capital usually invests in younger companies in innovative or technological industries as minority shareholders. Buyouts target larger companies in more mature industries and encompass Growth capital, Transmission Capital, and Distressed Capital depending on the company stage and needs. Leveraged buyouts (LBO), most common in Transmission Capital, are specific deals in which a small share of equity is invested and leveraged by a large acquisition debt.

The surge of Private Equity financing first occurred in the United States over the 1980s, a decade of intense corporate restructuring in the face of international competition and deregulation (Jensen 1993). It was driven by LBO and often relying on junk bond financing. As the junk bond market crashed, many LBO defaulted and investee companies went to bankruptcy, so that the Private Equity industry nearly disappeared in the early 1990s (Kaplan and Strömberg 2009). However, remaining Private Equity funds acted as a substitute for weak capital markets over the 1994–2004 period (Boucly et al. 2009) and the industry underwent a steady growth. In 2001, venture capital investments were almost 100 times larger than they were in 1977, whereas bank lending to small firms stayed constant at the best over the same period (Ueda 2004). The 2000s thus witnessed a new boom of the Private Equity and LBO peaked in the 2006–2007 due to a period of euphoric credit markets (Kaplan and Strömberg 2009).

The 2008 financial crisis led fundraising and debt markets to plummet and ended the Private Equity surge (Fig. 1). From then on, competition rocketed up and returns had to increasingly come from value creation, such as selecting underdeveloped companies and accelerating their growth, rather than financial leverage effect (Boucly et al. 2009). Price competition and consequent quest for value creation had nevertheless started prior to the crisis, as shown by Jin and Wang (2002) and Gaspar (2009). This article conjectures that responsible investing practices are part of this value creation quest.

Integration of Socially Responsible Practices by Private Equity Investors

The official intersection of Private Equity and SRI can be dated back to 2009, a period of liquidity dearth. Indeed, the

² Note that while General Partners in Private Equity funds have some discretion in how they value their portfolio, valuation is still subject to the fiduciary duties the general partner owes the fund.

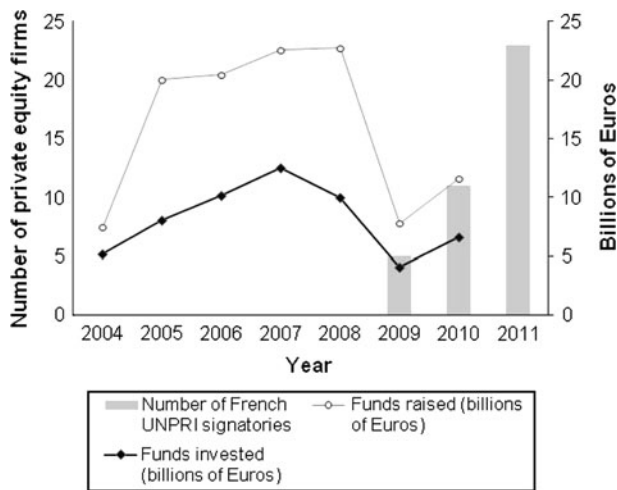


Fig. 1 The French Private Equity market (data AFIC 2011 and UNPRI 2011) in amounts of funds raised, funds invested, and UNPRI signatories

first major move toward responsible investment in Private Equity was the adoption (February 10, 2009) of guidelines covering environmental, health, safety, labor, governance, and social issues by large conventional players, namely the United States Private Equity Council, representing 13 of the world's largest Private Equity investors. Within a month, the Private Equity United Nations Principles for Responsible Investing (UNPRI) were launched and its Board stated (March 2009):

We also need to recognize that investors can, and should, be part of the response to this crisis and that responsible investment has an important role in mitigating future such market failures(...). This crisis demonstrated a clear lack of understanding of our underlying investments, and how they may have been putting not only the companies involved at risk, but the entire economy.

Between February 2009 and August 2011, 107 Private Equity companies became UNPRI signatories, among which world leader funds of funds and General Partners. National associations are major actors in this change, as witnessed by the Australian Private Equity and Venture Capital Association asking its members to “seriously consider signing the UN PRI” in August 2009 or the responsible investing guidelines provided in June 2010 by the British Association (BVCA 2010). In France, the Sustainable Development Project of the French Association for Private Equity (AFIC) was launched in April 2010 by conventional Private Equity firms. Key figures also encourage the SRI movement in Private Equity, such as Lord Mandelson, the United Kingdom secretary of state for business, innovation and skills, who emphasized that

embracing CSR issues into Private Equity investment approach made “good business sense” (September 2009).

Henceforth, SRI in Private Equity can be conjectured as a mainstream approach initiated by large players and structurally neither thought of nor developed as a niche market. Typically, socially responsible Private Equity hence essentially differs from conventional Private Equity in that it integrates ESG issues in its investing practices.

Let us finally note that despite its quick evolution, SRI in Private Equity is still at its first steps. The UNPRI estimate that the share of total Private Equity market subject to integration by PRI signatories worldwide was 5 % in 2009 and 8 % in 2011. As about 10 % of the AFIC members are UNPRI signatories, France appears as an active and interesting field to investigate in details the SRI movement in Private Equity.

Testable Hypotheses on Characteristics and Drivers of the Responsible Investment Movement in Private Equity

Does responsible Private Equity present specificities compared to SRI on financial markets? This section builds testable hypotheses on the characteristics of socially responsible investing in Private Equity and, respectively, its strategic and responsive drivers. Based on previous section, we here define socially responsible investing in Private Equity as the integration of ESG concerns in conventional investing practices.

Hypotheses on the Characteristics of Socially Responsible Private Equity

A straightforward conjecture is that difference between SRI on financial markets and socially responsible Private Equity might directly arise from the specificities of Private Equity itself. Drawing from the corporate finance literature, four characteristics of Private Equity are hence here detailed: information asymmetry reduction, agency costs cut, governance engineering, and operational engineering. Impacts on ESG criteria integration are conjectured and lead to testable hypotheses.

First, asymmetric information between informed managers and the public market has been shown to cause under-investment (Myers and Majluf 1984). Private Equity investors reduce this information asymmetry by monitoring the companies they select (Holmstrom and Tirole 1997) and evaluating them better than a standard financial institution would (Ueda 2004). As such, ESG integration appears as another tool to reduce information asymmetry and improve business.

Second, the agency theory framework has been applied to the Private Equity setting by Jensen (1986, 1989), leading to the free cash flow hypothesis. Jensen argues that managers in companies with excess cash flows have an incentive to waste organizational resources on personal ends, rather than pay out the excess cash to shareholders through dividends. However, financing by equity investors lead to high debt level (particularly for leverage buyouts) which affect the free cash flows of the firm for debt servicing, preventing opportunistic behaviors of managers (see, e.g., Desbrières and Schatt 2002). Private Equity tight hand on agency costs will naturally lead investors to carefully consider CSR in their portfolio companies, to avoid managerial entrenchment strategies (Baron et al. 2011; Cespa and Cestone 2007). Being a socially responsible Private Equity investor hence means fostering CSR in portfolio companies while reducing agency costs.

Both arguments foster the following hypothesis:

H1 Socially responsible Private Equity firms tend to integrate ESG issues in mainstream business, rather than considering it as a niche market segment.

Third, governance engineering has been shown to be another Private Equity strength (Kaplan and Strömberg 2009). Private Equity investors usually have significant impact (if not complete control) on portfolio company board and are much more involved in governance than public companies board. For instance, it is not unusual that they initiate a management change. Managing governance issues is thus already core in Private Equity business. Hereby, the gap between managing “G” to full “ESG” issues is more easily filled in by Private Equity investors than public investors.

H2a Private Equity firms are more likely to favor governance aspects within the ESG issues.

Forth, Kaplan and Strömberg (2009) argue that a property of Private Equity investors is operational engineering. Typically, Private Equity firms possess industry and operational expertise by their choice of human capital. They are thus able to advise companies at the core of their business to enhance value creation. Distinguishing value-creation CSR from managerial entrenchment requires a sound understanding of the firm environment and markets, which Private Equity possesses. Engagement thus appears a necessity to consistently be a socially responsible Private Equity investor. The previous hypothesis can thus be completed by:

H2b Socially responsible Private Equity heavily relies on investor engagement.

Understanding CSR potentiality needs sound expertise. Indeed, CSR encompasses many dimensions, with various

effects on financial performance (Brammer et al. 2006). Yet expertise on strategic CSR is still scarce, all the more considering the recent interest of the industry for ESG issues. It is thus more likely for large Private Equity firms benefitting from wide human capital resources to be able to appropriately acquire CSR expertise. This firm size effect might also be linked to a form of mainstreaming.

H3 The integration of ESG issues is more likely to be implemented in large Private Equity firms in terms of workforce.

Hypotheses on Strategic Drivers of Socially Responsible Private Equity

The idea of strategic ESG integration refers to the reconciliation of social and long-term economic interests of corporations. Experts in profit maximization and agency costs reduction, Private Equity investors would opportunistically analyze CSR as a strategic resource to improve the bottom line performance of companies (McWilliams et al. 2006). In their survey, Cumming and Johan (2007) indeed find that socially responsible institutional investment programs are more common among investors expecting greater economic returns from those investments. The following potential strategic drivers for responsible Private Equity are identified: value-creation, risk management, new market creation, differentiation, and related compliance to Limited Partners demand.

Hypothesis on Value Creation as a Strategic Driver

Can a Private Equity fund create value by integrating ESG criteria in investment practices?

As a starting point, empirical comparisons of public SRI performance to standard portfolios lead to rather lukewarm results, the former either achieving similar or lower yields than the latter (Bauer et al. 2005; Kreander et al. 2005).³

However, as previously underlined, Private Equity investors structurally differ from public investors. Beyond governance and operating engineering, they typically do not build large and diversified portfolios based on modern portfolio theory but rather select and follow a few companies in which they are significant active shareholders over a long-term horizon. More specifically, the Private Equity investment cycle consists in: (i) an acquisition stage over which firms can be screened on ESG issues; (ii) a 4–7 years of holding stage over which firms are monitored

³ Recent work suggests that whereas SRI underwent a learning period in which it underperformed standard portfolio, it since caught up with conventional funds performance (Climent and Soriano 2011).

and value can be created; and (iii), an exit stage where investors can benefit from the created value. Value creation in the Private Equity context should thus be analyzed through the lens of the link between CSR and financial performance, rather than SRI versus standards portfolios.

The link between CSR and performance has elicited much interest over the last three decades. Recent literature reviews (Orlitzky et al. 2003; Portney 2008; Reinhardt et al. 2008; Margolis et al. 2009) converge to a consensus on the absence of a financial penalty associated with CSR. Yet generating over-performance with proactive CSR does not come straightforward, leading academics to advocate research on how corporations can succeed in both performing on social and financial levels. Literature reviews also highlight that value can be created through different channels: increase input–output efficiency to get a competitive advantage (Derwall et al. 2005); reduction of costly employee turnover and the recruitment of motivated, hence more productive, employees (Turban and Greening 1997; Brekke and Nyborg 2008); and answer to consumers' demand and increase companies reputation and consumer loyalty, leading to long-term increased brand and company values (Kanter 1999; Porter and Kramer 2002; Kotler and Lee 2005).

The following hypothesis is thus proposed.

H4 Socially responsible investing in Private Equity is motivated by creating value in the companies owned.

Hypothesis on Risk Management as a Strategic Driver

Another strategic rationale for promoting CSR in companies is improved risk management. As the company's cash flow is used to service the acquisition debt, Private Equity investors' greater risk is that the company will not achieve the cash flow necessary (Le Nadant and Perdreau 2006). This is particularly crucial in the case of LBO transactions, in which investors are hence most interested in the company's future capacity to generate large and steady levels of cash flow. Scholtens (2006) identifies three specific risks particularly associated with ESG management default. A first direct risk is present when the investor takes possession of collateral, would it, for instance, poses a threat to the environment, as fines, decontamination, or compliance to standards can become costly. A second indirect risk arises from changes in environmental or social legislation (or consumer preferences), which can affect company's revenues and thereby its default probability. Finally, a reputational risk is also present where actions of borrowers may negatively feedback on its financiers. Hence, the following hypothesis is proposed.

H5 Socially responsible investing in Private Equity is a risk-management tool.

Hypothesis on New Markets Creation as a Strategic Driver

Environmental issues can create new business opportunities and offers wide innovation possibilities. This opening up of previously undiscovered market has early been highlighted as a rationale for CSR (Porter and Van der Linde 1995; Porter and Kramer 2002; Vogel 2005). The Private Equity industry already seized the opportunity of exploiting this “market for virtue” (Vogel 2005), as witnessed by their increasing involvement in the renewable energy and clean technology markets. Structurally, part of the Private Equity business exists to finance and support new companies and new markets.

Indeed, business angels, seed capital, and more globally speaking venture capital provide equity to managers creating or developing businesses. Typically, venture capital is oriented toward innovative firms (health sector, biotechnologies, informatics, energy sector) that can exploit new market opportunities based on consumers' new demand. Conversely, buyouts usually target more mature markets with steady cash flows (services, consumer goods, chemistry, industry) in which CSR rather consists in a transversal approach than in the development of brand new products.⁴ The following hypothesis can thus be proposed.

H6 Venture capitalists are more likely than other Private Equity investors to develop specialized green funds in order to open new markets.

Hypotheses on Differentiation and Related Compliance to Limited Partners Demand as Strategic Drivers

Differentiation can drive responsible investment in Private Equity firms under a twofold motivation: reducing competition intensity and capturing Limited Partners' preferences.

Competition intensity between Private Equity firms rocketed up with the financial crisis and the crash of equity raised worldwide (in France, equity raised dropped by 71 % in 2009). and competition for fundraising will keep on increasing over the next years as a consequence of the Basel III global regulatory standard on bank capital adequacy and liquidity, which ranked Private Equity funds as being among the most risky and illiquid assets.

⁴ However, cleantech funds could sometimes differ from responsible investment “per se”. For instance, investing in solar panels might not necessarily imply that a full ESG risk analysis has been conducted. Vogel (2005, p. 3), hence, points out that if socially responsible firms can benefit from green markets, “there is also a large place for their less responsible competitors.” The gap between green funds and socially responsible investing could also be suggested at the industry level by the co-existence in the French Private Equity Association of a “Sustainable Growth Club” and a “Green Techs Club”.

However, competition is likely to be heterogeneously perceived by Private Equity funds depending on whether they are *captive funds* or *independent funds*. A captive fund is significantly owned by its Limited Partners (typically a bank or a corporation); hereby fundraising is eased. On the contrary, an independent fund is owned by the Private Equity firm partners and needs to compete to raise funds. Previous literature already highlighted different behaviors (rather than mimetic processes) between captive and independent funds (Gompers and Lerner 2000; Hellmann 2002; Hellman et al. 2008)

Integrating ESG criteria might thus be used by Private Equity firms as a differentiation tool to attract Limited Partners.⁵ As fundraising competition is likely to be tougher for independent funds than captive funds, they might also be more induced to differentiate by offering CSR attributes to their investors.

H7a Independent funds are more likely than captive funds to integrate ESG issues as a differentiation tool to attract investors.

Another related motivation for CSR differentiation in the Private Equity context might also be to capture Limited Partners' preference. Theoretical and empirical work previously found that individual investors may derive non-financial utility from investing in SRI funds (Bollen 2007; Renneboog et al. 2008). Moreover, different investors exhibit differences in solvency and returns requirements, extent of regulatory oversight, corporate objectives, and accounting rules (Cumming and Johan 2007). They might also differ in their CSR commitments (Cox et al. 2004; Johnson and Greening 1999; Waddock and Graves 1994). Among long-term investors, the regulatory constraints which European Pension funds already face (see Renneboog et al. (2008)' review of Pension funds European regulations) are likely to increase their interest for French responsible investments. The following hypothesis is thus proposed.

H7b Funds with long-term investors, in particular pension-funds, as Limited Partners are more likely to integrate ESG issues.

Hypothesis on Responsive Drivers of Socially Responsible Private Equity

Strategic drivers of responsible investing are typically opposed to responsive drivers. Whereas the former imply a

proactive use of CSR to generate profits, the latter hints that sacrificing profits to provide environmental or social performance will hinder the firm from social pressure. Supporting this dichotomy, Baron et al. (2011) empirically show that responsive CSR increases with the firm's slack resources and not strategic CSR.

The Private Equity industry has been the focus of intense criticism over the past few years, both in Anglo-Saxon and Continental Europe countries, which was blamed for short-termism impacting employees, drying-off of investing capacities, and prevalence of short-term financial profits over long-term industrial growth in companies financed by the Private Equity industry. Yet, evidence that non-financial stakeholders suffer Private Equity short-termism is thus at best mixed (Boucly et al. 2009; Davis et al. 2011; Amess and Wright 2007).

However, the financial crash the world economy underwent drew public attention to the funds' activity. The Private Equity recent surge also made it clearly visible. A General Partner hence stated in the British Venture Capital Association report (2010): "General Partners must accept that, now that they are managing large amounts of money, they face increased scrutiny from governments." Grounded or not, the reputation of Private Equity has been tarnished and its social interest is now contested.

Cases of visible polluting industries increasing their environmental and/or social performance under social pressure are many in the CSR literature. The "license to operate" concept proposed by Post et al. (2002) well summarizes that a firm can hardly survive long when its social interest is contested and its actions have lost legitimacy.

However, social pressure is likely to focus on specific Private Equity actors, namely large visible firms and/or LBO. Indeed, Cumming and Johan (2007) indicate that socially responsible Private Equity investment programs are more common among larger institutional investors. This finding is in line with larger companies tending to be more scrutinized than smaller ones. Second, LBO have been more subject to social criticism than other types of Private Equity, even though Venture Capital's drawbacks have also been pointed out (Hellmann et al. 2008; Ueda 2004). LBO and Transmission funds being both larger and more subject to criticism than other Private Equity funds types, the following hypothesis can be proposed.

H8 Large and thus visible Private Equity firms as well as LBO specialists are more likely to formalize socially responsible investing and to publicly communicated it to protect their reputation and license-to-operate.

The following section now presents the data on which the established hypotheses are tested.

⁵ CSR has already been shown to be a means of differentiation in otherwise competitive environments (Arora and Gangopadhyay 1995; Fisman et al. 2007), and to most strongly affect performance in low-innovation firms and in industries little segmented (Hull and Rothenberg 2008), as is the case of the Private Equity industry.

Data and Method

The French Private Equity Industry

Second largest market in Europe after the United Kingdom, the French Private Equity market (comprising both venture capital and buyouts) is also the third market worldwide in amounts invested behind the United States market (AFIC 2010).

The French Private Equity industry is usually segmented in: seed and venture capital (9.4 % funds invested), growth capital (35.9 %), and transmission capital transactions (54.7 %; data AFIC, 2010). LBO mainly occur in Transmission Capital and essentially deal with divestments of subsidiaries within groups (“spin-offs”; 64.1 %); transmission of family businesses (30.8 %) and to a minor extent with stockmarket-listed companies going private (“PTPs”; 5.1 %).

The French Private Equity market is hence representative of the continental European market.

However, some differences appear compared to Anglo-Saxon markets and are worth noticing. For instance, PTPs are limited in France compared to the United States and the United Kingdom industries due to regulatory constraints (Le Nadant and Perdreau 2006). Another specificity of the French Private Equity market arises from lower debt pressure (Desbrières and Schatt 2002) as well as the substantial part of their personal wealth French managers usually invest in the company’s capital (Desbrières and Schatt 2002).

Sampling and Structural Data

The sample gathers observations on 309 Private Equity firms in 2011, all located in France. It is thus close to the whole French industry, and include 278 out of the 280 members of the French Private Equity Associations, non-members of the Association and French local Private Equity firms sponsored by the French Sovereign Wealth Funds (FSI). We focus on the Private Equity firm level rather than on the Private Equity fund level. Hence, a firm in our sample usually managed several funds simultaneously.⁶

Equity Data were collected from the Private Equity firms’ website, press releases, and specialized media.⁷ Data

includes main characteristics such as the Private Equity firm age (*Firm Age*) and the number of funds it manages (*Funds*), which both give information on the firm experience (Cumming et al. 2009); as well as the number of companies financed by the firm at the time of data collection (*Companies*). Firm size is measured in terms of asset managed (*Assets managed*) and number of investors (*Workforce*). The firm activity is given by dummy variables indicating whether it engages in venture capital, transmission (buyouts), growth capital, mezzanine, distressed capital or funds of funds transactions.

Data are also collected on the ownership structure of the Private Equity firms, as it has been shown to impact the provision of capital (Cumming et al. 2008). We distinguish the number of shares owned by another Private Equity firm, individual owners–managers (*Partners*), banks, insurance companies, industrial corporations, and the French State (via government funds). Limited Partners are accounted for by dummy variables indicating whether the firm manages capital provided by an industrial corporation, a sovereign wealth fund, individual investors, pension funds, family offices, or if they are captive. A dummy variable also indicates whether the firm manages at least a fund specialized in sector industry. Management style is grasped by variables on the firm CEO gender and background. Geographical investment scope is also controlled for as a growing body of work suggests that international differences in legal and institutional factors can affect Private Equity and SRI allocations (e.g., Lerner and Schoar 2005; Cumming et al. 2010). In particular, dummy variables indicate whether the firms invest only in a specific French district, in European countries, or in non-European countries (“international scope”).

Data on public responsible investment practices is also gathered: signatory of the UNPRI and of the French National Association Ethics Chart; management of green or social funds; web communication on responsible investment practices; and interest as proxied by answer to responsible investment surveys.

Variable description and descriptive statistics can be found in Table 1 and the correlation matrix in Table 8 in the Appendix. Continuous variables were checked for outliers using the Grubbs (1969) test. To limit influential points, the logarithm of the variable “assets managed” was used in regressions.

Survey Data

Whereas structural and investment data can be collected in the public domain, such is not the case of strategies behind responsible investment practices. To get an insider perspective, the database is thus completed with survey data.

⁶ We are thus not able to control for style drift (see Cumming et al. 2009) that is the deviation from stated objectives of the funds. Indeed we focus on the Private Equity organization, not the Private Equity fund, and a Private Equity firm may comprise more than one fund and may thereby provide investors investment products across a range of stated investment styles.

⁷ Specialized media used to built the database include the 2010 and 2011 Private Equity Firms Guide (“Le Guide des Sociétés de Capital Investissement”) and the <http://investing.businessweek.com> website.

Table 1 Descriptive statistics on public data

Variable name	Definition	Observations	Mean	Standard deviation	Minimum	Maximum
Main characteristics						
Firm age	Years elapsed since firm foundation ^a	310	13.2	11.7	0	102
Workforce	Number of employees	286	13.2	27.2	1	371
Funds	Number of funds managed by the firm ^b	302	5.0	8.7	0	65
Companies	Number of companies hold in portfolio	288	39.7	74.2	0	700
Assets managed	Millions of Euros of assets managed ^c	295	876.2	2951.7	0	32000
Ownership: percentage of firms shares...						
Listed	... listed on a Stock market (in %)	303	0.01	0.09	0	1
PE firm	... owned by another Private Equity firm (in %)	274	0.16	0.34	0	1
Partners	... owned by the firm Partners (in %)	274	0.42	0.47	0	1
Bank	... owned by a Bank (in %)	274	0.18	0.33	0	1
Insurance company	... owned by an insurance company (in %)	274	0.03	0.16	0	1
Industry	... owned by an industrial corporation (in %)	274	0.05	0.19	0	1
French state	... owned directly or indirectly by the French State (in %)	274	0.11	0.26	0	1
Other	... owned by other shareholders (in %)	274	0.03	0.14	0	1
Limited Partners: dummy variable equal to 1 if..., 0 otherwise						
LP Industry	...the firm manages funds provided by an industrial corporation	271	0.19	0.39	0	1
LP Captive	... the firm is captive and essentially manages funds provided by a single bank or insurance	271	0.32	0.47	0	1
LP Sovereign	...the firm manages funds provided by a Sovereign Wealth Funds	274	0.39	0.49	0	1
LP Individuals	... the firm manages funds provided by individuals	273	0.21	0.41	0	1
LP Institutional	... the firm manages institutional funds (except pension funds)	267	0.76	0.43	0	1
LP Pension funds	... the firm manages funds provided by pension funds	267	0.26	0.44	0	1
LP Family offices	... the firm manages funds provided by family offices	266	0.37	0.48	0	1
Activity						
Venture Capital	Percentage of funds invested in venture capital	303	0.42	0.49	0	1
Transmission	Percentage of funds invested in transmission capital	303	0.60	0.49	0	1
Growth	Percentage of funds invested in growth capital	303	0.59	0.49	0	1
Mezzanine	Percentage of funds invested in mezzanine	303	0.07	0.26	0	1
Distressed capital	Percentage of funds invested in distressed capital	303	0.09	0.28	0	1
Funds of funds	Percentage of funds invested in funds of funds	303	0.11	0.31	0	1
Investment scope: dummy variable equal to 1 if..., 0 otherwise						
Minority	... the firm invests as a minority shareholder	242	0.78	0.42	0	1
Regional scope	... the firm invests only in a specific French district	311	0.92	0.27	0	1
European scope	... the firm invests in other European countries than France	311	0.58	0.50	0	1
International scope	... the firm invests in other countries than European countries	311	0.21	0.40	0	1
Sector	... the firm manages at least one fund specialized in a sector	302	0.36	0.48	0	1
Management: dummy variable equal to 1 if the Chairman..., 0 otherwise						
Gender	... is a woman	299	0.05	0.21	0	1
Founder	... founded the firm	290	0.43	0.50	0	1
Engineer	... has a French “Grandes Ecoles” Engineer background	256	0.28	0.45	0	1

Table 1 continued

Variable name	Definition	Observations	Mean	Standard deviation	Minimum	Maximum
Business school	...has a French Business School background	254	0.47	0.50	0	1
International	... has an international background	254	0.22	0.42	0	1
Other	... has a different French background	254	0.46	0.50	0	1
Public responsible investment practices: dummy variable equal to 1 if the firm..., 0 otherwise						
UNPRI	... is a UNPRI signatory	316	0.09	0.28	0	1
AFIC chart	... is signatory of the French Private Equity Association Ethics Chart	317	0.67	0.47	0	1
Green or Social fund	... manages at least one fund with a social or environmental target	302	0.12	0.32	0	1
Communication	...has a website referring to responsible investing practices	297	0.18	0.38	0	1
Interest	... answered to responsible investment survey	308	0.24	0.43	0	1

This table summarizes the public data on the characteristics of French Private Equity firms

^a The age of firms founded in 2011 has been rounded to 0

^b Depending on their legal form, some firms do not manage “funds” as legally defined

^c Two firms were in the process of raising fund and did not managed assets in 2011

The survey was built in partnership with Novethic,⁸ a subsidiary of the French public institution and long-term investor Caisse des Dépôts et Consignations, and sent to 308 Private Equity firms in March 2011. The questionnaire comprises items on practical ESG integration and formalization, dedicated resources, greenhouse gas emission assessment, positive or negative screening, support provided to the company portfolio to foster CSR, belief in the importance of ESG integration for Limited Partners and firm reputation. Questionnaires were emailed and filled online. Descriptive statistics can be found in Table 2.

We obtained a response rate of 24.0 % (74 respondents out of 308 firms). This rate compares favorably with those observed in previous surveys on responsible investing in Private Equity: 7 % (100 answers out of 1114 Dutch institutions) in Cumming and Johan (2007), 20 % (84 answers out of 415 UK firms) in BVCA (2009). In a survey on Private Equity deal structures in Italy, Cumming and Zambelli (2010, 2012) obtained a 47 % response rate (27 out of 57 funds) and note that rates observed in previous surveys in finance range from 19 % in the United States (Brau and Fawcett 2006) to 60 % in the United Kingdom (Wright and Robbie 1996).

To limit social desirability bias, the questionnaire made no reference to “socially responsible investment” or to “ethics” (Neumann 2003; Vyvyan et al. 2012). To stimulate frank answers, the anonymity of respondents was

ensured (Kuckertz and Wagner 2009). Questionnaire wording (translated from French)⁹ is presented in the Appendix.

Another common bias in surveys arises from self-exclusion of respondents. Following Moore and Reichert (1983) and Kuckertz and Wagner (2009), the issue of the survey sample representativeness is dealt with by comparing firm characteristics of the respondents to the non-respondents. As our sample covers the whole French Private Equity industry, this comparison provides us with a robust test of our survey representativeness. Univariate tests were used (*t* tests for normally distributed variables; test of proportions for categorical data). Detailed results can be found in Table 9 in the Appendix.

The conducted tests lead us to conclude that there is no statistical difference between responding and non-responding groups in terms of firm age, structure of ownership, geographical investment scope, and industry sector investment scope. Little bias is observed in terms of spread between activities (venture capital, growth capital, mezzanine, funds of funds, reversal, and LBO, the latter being slightly overrepresented) and between Limited Partners categories (funds with institutional Limited Partners being slightly overrepresented). However, as expected (Kuckertz and Wagner 2009), replies contain a disproportionate number of companies particularly concerned about responsible investing as shown by the significantly larger proportions of UNPRI and AFIC Ethic Chart signatories, firms communicating about CSR on their website; and firms managing green or social funds. Hereby, survey answers will be used in further regressions as a proxy for

⁸ Our research was conducted in full independence and without any biases for the best representativeness to be obtained. In fact, Novethic granted access to their data and did not interfere in our study, nor used our results (they conducted independent descriptive research on this topic, not academic one).

⁹ The original French questionnaire is available upon request.

Table 2 Descriptive statistics of survey data

	All respondents			UNPRI signatories	
	Number of observations	Mean	Standard deviations	Number of observations	Mean
Formalization					
ESG Chart	74	0.35	0.48	22	0.59
ESG Formation of PE firm employees	74	0.46	0.50	22	0.73
ESG Issues managed by a specialized employee	69	0.20	0.40	22	0.36
ESG Issues managed by a non-specialized employee	69	0.35	0.48	22	0.36
ESG Issues directly managed by investors	69	0.26	0.44	22	0.23
ESG Issues managed by a specialized third party	69	0.03	0.16	22	0.05
Engagement					
ESG Chart sent to portfolio companies	74	0.26	0.42	22	0.36
Monitor E. performance of portfolio companies	74	0.20	0.40	22	0.41
Monitor S. performance of portfolio companies	74	0.53	0.50	22	0.64
Monitor G. performance of portfolio companies	74	0.66	0.48	22	0.68
Demand ESG reporting to portfolio companies	74	0.22	0.41	22	0.41
ESG Issues brought to company's supervisory board	74	0.23	0.42	22	0.32
ESG Issues in shareholders' pact	74	0.18	0.38	22	0.18
Visit companies and plants	74	0.26	0.44	22	0.27
Screening					
Sectoral or normative exclusion	74	0.84	0.37	22	0.73
Already discarded investment on ESG grounds	74	0.45	0.50	22	0.59
Value financial performance driver					
ESG Impact measured	66	0.17	0.38	21	0.29
Already bargained lower company price on ESG grounds	74	0.12	0.33	22	0.18
ESG Issues matter for risk management	74	0.64	0.48	22	0.77
ESG Issues matter to increase company value	74	0.43	0.50	22	0.54
LP demand driver					
ESG Chart communicated to Limited Partners	74	0.24	0.43	22	0.41
ESG Reporting communicated to Limited Partners	74	0.12	0.33	22	0.32
ESG Performance matters to Limited Partners	74	0.51	0.50	22	0.73
Reputation driver					
ESG Chart public	74	0.05	0.23	22	0.18
ESG Reporting public	74	0.01	0.12	22	0.05
ESG Issues matter for reputation and image risk	74	0.46	0.50	22	0.59

This table presents summary statistics of the survey data collected. Statistics are given for all respondents and for the sub-sample of respondents which also are signatories of the United Nations Principles for Responsible Investing

firm's interest in responsible investment practices. Respondents are also significantly larger than non-respondents in terms of assets managed, workforce, number of funds and companies in portfolio. We explain the firm size bias as being related to the overrepresentation of firms interested in responsible investing practices, as discussed in sections "[Hypotheses on the characteristics of socially responsible Private Equity](#)" and "[Hypothesis on responsive drivers of socially responsible Private Equity](#)." Nonetheless, the sample still includes a wide range of firm size (from 0.7 to 28,000 millions of Euros managed; median 318 millions of Euros). Those elements lead us to conclude

that our survey sample well represents the French Private Equity industry.

Multivariate Empirical Analysis

Descriptive statistics on public and survey data provide first elements of analysis (Tables 1, 2). Probit regressions are then used to analyze the factors driving the probability that a Private Equity firm implements responsible investment practices. Three public indicators of responsible investing are empirically investigated: being a UNPRI

Table 3 Multivariate analysis (probit) on UNPRI signature drivers

	Model 1		Model 2	
	Marginal effect	z-stat.	Marginal effect	z-stat.
Firm characteristics				
Age	-1.95e-05	-0.06	5.53e-06	0.02
Workforce			5.76e-04	2.18**
Companies	-9.09e-05	-1.33	-1.49e-04	-1.94*
Log(Assets managed)	8.33e-03	2.10**		
Ownership				
PE firm	4.44e-02	1.47	-4.85e-03	-0.40
Partners	4.78e-02	1.67*		
Bank	4.21e-02	1.28		
Insurance company	4.71e-02	1.09	1.66e-02	0.71
Industry	6.11e-02	1.77*		
French state	4.87e-02	1.56		
Limited Partners				
LP Industry			-8.87e-03	-1.37
LP Captive			-4.79e-03	-0.56
LP Sovereign			1.39e-02	1.03
LP Individuals	1.98e-02	1.19	4.52e-03	0.40
LP Institutional	4.98e-03	1.15	3.20e-03	0.85
LP Pension funds	1.03e-02	0.94	7.55e-03	0.77
LP Family offices	-1.23e-03	-0.15	-1.15e-02	-1.45
Activity				
Venture Capital	9.43e-03	0.74	4.54e-03	0.42
Transmission	-6.27e-03	-0.52	1.15e-03	0.12
Growth	-1.97e-02	-1.34	-2.68e-02	-1.65
Funds of funds	3.71e-02	0.85	3.40e-03	0.17
Mezzanine	2.27e-01	1.94*	2.11e-01	1.90*
Investment scope				
Minority	1.90e-02	2.02**	1.97e-02	2.07**
Regional scope	5.97e-03	0.28	-5.89e-04	-0.04
European scope	-3.36e-02	-1.61	-9.30e-03	-0.77
International scope	-6.66e-03	-0.91	-8.37e-03	-1.36
Sector	1.22e-03	0.13	1.30e-02	1.12
Management				
Founder			4.76e-03	0.62
Engineer	-9.16e-03	-1.24	-8.09e-03	-1.14
Business school	-1.29e-02	-1.35	-1.03e-03	-0.12
International	1.82e-02	1.11	1.35e-02	0.98
Responsible investment practices				
AFIC Chart	2.17e-02	1.84	2.13e-02	1.98**
Green or Social fund	-7.70e-03	-1.35	-7.50e-03	-1.45
Communication	1.08e-01	2.85***	9.68e-02	2.64***
Interest	1.44e-01	3.45***	1.69e-01	3.51***
Nb. of obs.	166		163	
Nb. of obs. where = 1	21		21	
Pseudo- R^2	53.08		52.67	
Loglikelihood function (L)	-29.57		-29.64	
LR- χ^2 statistics	66.91***		65.97***	

Table presents the results obtained by fitting maximum-likelihood probit models (model 1 and model 2). It reports marginal effects on the change in the probability of becoming a United Nations Principles for Responsible Investing (UNPRI) signatory (change in the probability for an infinitesimal change in each continuous variable; discrete change in the probability for dummy variables). Pseudo- R^2 is calculated by McFadden's formula, where pseudo- $R^2 = 1 - \log L / \log (LR)$. The LR- χ^2 statistics provides a test of the model robustness by indicating that the model as a whole fits significantly better than a model with no predictors. Two-tailed tests were used. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4 Multivariant analysis (probit) on UNPRI signature drivers

	Model 3		Model 4	
	Marginal effect	z-stat.	Marginal effect	z-stat.
Firm characteristics				
Age				
Workforce				
Companies				
Log(Assets managed)	1.84e−02	3.23***	1.98e−02	2.86***
Ownership				
PE firm	1.40e−01	1.78*		
Partners	1.27e−01	1.69*		
Bank	1.02e−01	1.30		
Insurance company	6.09e−02	0.72		
Industry	1.44e−01	1.71*		
French state	1.44e−01	1.79*		
Limited Partners				
LP Industry				
LP Captive				
LP Sovereign				
LP Individuals				
LP Institutional				
LP Pension funds				
LP Family offices				
Activity				
Venture Capital			4.73e−03	0.23
Transmission			−1.05e−02	−0.46
Growth			9.89e−03	0.50
Funds of funds			9.74e−03	0.29
Mezzanine			6.01e−03	0.17
Investment scope				
Minority				
Regional scope			−9.57e−03	−0.26
European scope			−1.03e−02	−0.40
International scope			−3.67e−02	−2.09**
Sector				
Management				
Founder				
Engineer				
Business school				
International				
Responsible investment practices				
AFIC Chart	5.11e−02	2.05**	5.39e−02	2.11**
Green or Social fund	−1.44e−03	−0.05	−1.76e−02	−0.68
Communication	1.56e−01	3.69***	1.97e−01	4.10***
Interest				
Nb. of obs.	257		281	
Nb. of obs. where = 1	24		26	
Pseudo- R^2	30.89		31.18	
Loglikelihood function	−55.11		−59.63	
LR- χ^2 statistics (L)	49.27***		54.03***	

Table presents the results obtained by fitting maximum-likelihood probit models (model 3 and model 4). It reports marginal effects on the change in the probability of becoming a United Nations Principles for Responsible Investing (UNPRI) signatory (change in the probability for an infinitesimal change in each continuous variable; discrete change in the probability for dummy variables). Pseudo- R^2 is calculated by McFadden's formula, where pseudo- $R^2 = 1 - \log L / \log (LR)$. The LR- χ^2 statistics provides a test of the model robustness by indicating that the model as a whole fits significantly better than a model with no predictors. Two-tailed tests were used. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

signatory (Tables 3, 4), managing a funds specialized on environmental or social issues (Table 5), and communicating on responsible investing on website (Table 6). Using those three alternative measures of responsible investing enables us to check whether they have similar underlying drivers. For all the three dependent variable, we also provide four regression models (“model 1” to “model 4” in Table 3, 4, 5, 6) to show result robustness to alternative subsets in the sample and alternative explanatory variables (as in Cumming and Johan 2007).

Results are now presented and discussed.

Results on Characteristics of Socially Responsible Private Equity

Results confirm hypothesis H1: Responsible Private Equity firms do not belong to an SRI niche market but rather integrate ESG issues in mainstream business. Indeed, we find that 12 % firms manage at least one environmental or social funds (Table 1). Yet only six firms (<2 %) specialize on such funds. Most of those who do manage green funds also manage classic funds. Signatories of the UNPRI and firms who answered survey, i.e., firms who have an interest for CSR issues, are widely conventional Private Equity funds.

Results also confirm hypothesis H2a: Private Equity firms are more likely to favor governance aspects within the ESG issues. Table 2 displays that governance was monitored by 66 % responders, significantly exceeding Environment (20 %) and Social (53 %) issues.

Results support hypothesis H2b: Socially responsible Private Equity investors heavily relies on investor engagement. Indeed, we see from Table 2 that 23 % brought ESG issues to company’s supervisory board, 53 % responders used direct monitoring of social issues in company (64 % among UNPRI signatories), 26 % visited companies or plants, and 22 % demanded ESG reporting to portfolio companies (41 % for UNPRI). However, ESG issue management is not legally enforced though the shareholders’ pact. We can note that engagement is also associated with ESG screening. 84 % firms already discarded an investment opportunity of ESG grounds (see Table 2). Let us note that this high percentage might be driven up by governance issues, at the core of the Private Equity business.

Finally, results partly confirm hypothesis H3: socially responsible investing is more likely in large Private Equity firms in terms of workforce. Indeed, the probability of being a UNPRI signatory (Table 3) increases with workforce size, yet in a small extent. Precisely, the probability of being a UNPRI signatory increases by 0.058 % for any additional employee (model 2, Table 3, $p < 0.05$). Moreover, survey descriptive statistics indicate that 46 % of survey respondents provide and ESG formation to their

employees (Table 2). This formation argues both in favor of the current lack of human capital to manage such issues, which might be problematic for ESG management, as well as a real involvement to develop this human capital.

Results on Strategic Drivers of Socially Responsible Private Equity

Results partly support hypothesis H4: Socially responsible investing in Private Equity are motivated by creating value in the companies owned, yet not only. Indeed, survey data highlight that 43 % of respondents only agree with ESG issues being important to create value (54 % among UNPRI signatories; Table 2). We can note that only 12 % ever measured ESG impact on their portfolio performance. Hence, most investors believing that responsible investing creates value do so on qualitative grounds.

Conversely, hypothesis H5 is confirmed by survey answers: socially responsible investing in Private Equity are a risk-management tool. 64 % of the respondents (77 % of UNPRI signatories) believe ESG issues matter for risk management (Table 2).

Strong support is also brought to hypothesis H6: Venture capitalists are more likely than other Private Equity investors to develop specialized green funds to open new markets. The probability to manage such funds statistically significantly increases when the firm is a venture capitalist and decreases in transmission capital (buyouts). Precisely, marginal effects reported in Table 5 indicate that the probability of managing a green or social fund is 5.55 % higher (model 4, $p < 0.05$) in venture capital funds and 7.36 % lower in transmission capital ($p < 0.05$). We can also note that Growth Capital firms are also very active on GreenTechs. Their activity is indeed more in line with venture capital than with buyouts as they invest in expansion cycles of companies. Consequently, Private Equity firms managing green funds hence show typical characteristic of venture capitalists, as significantly smaller firms in terms of workforce and assets. Probit results also confirm the dichotomy between specialized green funds and responsible investing. Indeed, being a UNPRI signatory does not increase the probability to manage a green fund (Table 5), and, respectively, managing a green funds does not increase the probability to be a UNPRI signatory (Tables 3, 4).

Evidence is mixed on hypothesis H7. As a primer, the belief among General Partners that ESG issues matter for Limited Partners is strongly supported by survey data. 51 % of the respondents agree so, up to 73 % among UNPRI signatories. A few firms (12 %) actually report their ESG performance to Limited Partners, most likely because they are themselves still implementing ESG reporting at their portfolio level.

Table 5 Multivariate analysis (probit) on green or social funds management drivers

	Model 1		Model 2		Model 3		Model 4	
	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.
Firm characteristics								
Age	5.11e−08	1.92*	4.41e−11	1.71*				
Workforce			−6.53e−11	−1.65*				
Log(Assets managed)	−2.19e−07	−1.11			−1.91e−02	−2.12**	−9.13e−03	−1.12
Ownership								
PE firm	−3.40e−07	−0.20	−2.07e−09	−1.47	−7.60e−04	−0.01		
Partners	−3.10e−08	−0.04			−2.51e−02	−0.31		
Bank	9.63e−07	0.93			1.14e−02	0.13		
Industry	4.12e−06	1.90*			7.94e−02	0.78		
French state	−3.26e−06	−1.76*			6.60e−02	0.70		
Limited Partners								
LP Industry			6.70e−07	1.22				
LP Captive			1.13e−11	0.02				
LP Sovereign			3.32e−10	0.42				
LP Individuals	−1.69e−07	−0.34	1.47e−05	1.55				
LP Institutional	5.99e−07	1.66*	6.69e−10	1.46				
LP Pension funds	4.86e−04	1.83*	1.64e−07	1.46				
LP Family offices	−1.52e−07	−0.24	1.60e−09	0.52				
Activity								
Venture Capital	3.24e−05	1.99**	1.56e−06	1.72*			5.55e−02	1.99**
Transmission	−7.88e−03	−2.16***	−1.80e−04	−1.72			−7.36e−02	−2.35**
Growth	7.63e−05	2.08**	1.49e−05	1.46			7.66e−02	2.68***
Funds of funds	4.04e−03	1.56	3.18e−05	1.34			1.85e−02	0.38
Investment scope								
Minority	−1.37e−04	−1.40	−2.75e−09	−0.62				
Regional scope	−3.49e−07	−0.71	−1.06e−09	−0.96			3.87e−02	0.84
European scope	−4.87e−05	−1.37	−1.09e−07	−1.09			−1.63e−03	−0.05
International scope	−1.85e−07	−0.45	−1.17e−10	−0.55			2.21e−02	0.64
Management								
Founder			−5.30e−09	−1.00				
Engineer	9.95e−05	1.59	1.03e−07	1.44				
Business school	6.58e−09	0.01	−1.65e−10	−0.28				
International	2.05e−04	1.38	1.32e−06	1.70*				
Responsible investment practices								
AFIC Chart	−3.53e−07	−0.26	−7.03e−06	−1.36	6.45e−02	1.77*	2.82e−02	0.97
UNPRI	4.48e−08	0.05	5.10e−06	1.14	−3.57e−02	−0.80	−2.85e−02	−0.82
Communication	2.09e−02	2.18**	1.36e−03	2.00**	3.95e−01	5.16***	3.40e−01	5.40***
Interest	1.35e−05	1.27	2.27e−07	1.50				
Nb. of obs.	174		167		257		281	
Nb. of obs. where = 1	14		14		27		30	
Pseudo- R^2	65.03		65.97		24.25		32.56	
Loglikelihood function	−17.03		−16.37		−65.43		−64.38	
LR- χ^2 statistics	63.34***		63.47***		41.88***		62.15***	

Table presents results obtained by fitting maximum-likelihood probit models (model 1, model 2, model 3, and model 4). It reports marginal effects on the change in the probability of managing a Private Equity funds specialized on green or social issues (change in the probability for an infinitesimal change in each continuous variable, discrete change in the probability for dummy variables). Pseudo- R^2 is calculated by McFadden's formula, where pseudo- $R^2 = 1 - \log L / \log (LR)$. The LR- χ^2 statistics provides a test of the model robustness by indicating that the model as a whole fits significantly better than a model with no predictors. Two-tailed tests were used. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6 Multivariant analysis (probit) on CSR communication drivers

	Model 1		Model 2		Model 3		Model 4	
	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.
Firm characteristics								
Age	-6.38e-4	-0.31	2.19e-4	0.13				
Workforce			1.31e-3	1.06				
Log(Assets managed)	3.92e-3	0.18			3.53e-2	2.99***	2.65e-2	1.99**
Ownership								
PE firm	-3.31e-1	-2.21**	-1.04e-1	-1.23	-2.49e-1	-2.34**		
Partners	-1.91e-1	-1.49			-1.78e-1	-1.95*		
Bank	-3.23e-1	-2.16**			-2.82e-1	-2.58***		
Industry	-7.42e-3	-0.03*			-8.05e-2	-0.60		
French state	-3.00e-1	-1.78			-2.07e-1	-1.57		
Limited Partners								
LP Industry			-1.49e-3	-0.02				
LP Captive			1.74e-2	0.28				
LP Sovereign			4.23e-2	0.60				
LP Individuals	1.35e-1	1.70*	9.22e-2	1.36				
LP Institutional	-9.51e-2	-1.14	-9.25e-2	-1.31				
LP Pension funds	-3.45e-2	-0.55	-2.33e-2	-0.42				
LP Family offices	-8.54e-2	-1.41	-1.08e-1	-1.96**				
Activity								
Venture Capital	2.95e-2	0.42	-2.52e-2	-0.37			7.73e-3	0.17
Transmission	5.44e-2	0.72	5.96e-2	0.90			1.55e-2	0.35
Growth	-6.15e-2	-0.88	-8.54e-2	-1.28			-1.89e-2	-0.45
Mezzanine	-8.07e-2	-0.92	-5.88e-2	-0.76			-6.80e-2	-1.00
Funds of funds	-4.55e-2	-0.44	-3.23e-2	-0.37			-3.41e-2	-0.56
Investment scope								
Minority	-4.85e-2	-0.52	-5.77e-2	-0.72				
Regional scope	1.42e-1	0.97	6.78e-3	0.06			-1.18e-2	-0.15
European scope	1.95e-1	2.53**	1.70e-1	2.53**			6.75e-2	1.25
International scope	8.82e-2	1.13	8.92e-2	1.23			5.30e-2	1.00
Management								
Founder			8.25e-2	1.32				
Engineer	8.68e-2	1.25	9.57e-2	1.53				
Business school	-2.12e-2	-0.36	1.83e-2	0.34				
International	8.04e-2	1.06	7.51e-2	1.10				
Responsible investment practices								
AFIC Chart	3.31e-4	0.00	4.23e-2	0.75	6.94e-3	0.14	1.12e-2	0.24
UNPRI	3.17e-1	2.54**	2.10e-1	1.99**	2.88e-1	2.96***	3.15e-1	3.38
Green or social funds	6.48e-1	3.54***	7.02e-1	4.04***	5.17e-1	5.36***	5.19e-1	5.48***
Interest	4.18e-2	0.61	4.63e-2	0.72	4.04e-2	0.77	6.45e-2	1.25
Nb. of obs.	174		167		257		281	
Nb. of obs. where = 1	36		35		43		48	
Pseudo- R^2	45.34		46.16		33.62		33.51	
Loglikelihood function	-48.49		-46.16		-77.06		-85.42	

Table 6 continued

	Model 1		Model 2		Model 3		Model 4	
	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.	Marginal effect	z-stat.
LR- χ^2 statistics	80.44***		79.16***		78.03***		86.10***	

Table presents results obtained by fitting maximum-likelihood probit models (model 1, model 2, model 3, and model 4). It reports marginal effects on the change in the probability of the Private Equity firm to communicate on its website about corporate social responsibility, socially responsible investment or the management of ESG issues (change in the probability for an infinitesimal change in each continuous variable; discrete change in the probability for dummy variables). Pseudo- R^2 is calculated by McFadden's formula, where $\text{pseudo-}R^2 = 1 - \log L/\log (LR)$. The LR- χ^2 statistics provides a test of the model robustness by indicating that the model as a whole fits significantly better than a model with no predictors. Two-tailed tests were used. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Hypothesis H7a is strongly confirmed: Independent funds are more likely than captive funds to develop responsible practices as a differentiation tool to attract investors. Indeed, we observe that Private Equity firms owned by their partners, meaning independent, have a 12.7 % higher probability of being UNPRI signatories (model 3, Table 3, $p < 0.10$; effect of 4.78 % in model 3). However, no significant effect of specific categories of Limited Partners is found on the probability to be a UNPRI signatory, neither positive nor negative. In particular, Captive funds are not less likely to be signatories. Results also show that Private Equity firms owned by banks (bank captive funds) are less likely to communicate on responsible investing on their website than other firms. Precisely, marginal effects show that they have a 33 % lower probability (model 1, Table 6, $p < 0.05$, effect of 25 % in model 3). Similarly, Private Equity firms owned by other Private Equity firms (Private Equity captive funds) also have a 32 % lower probability (model 1, Table 6, $p < 0.05$; effect of 28 % in model 3). This finding is in line with Cumming et al. (2008), who find in the context of Japanese venture capitalists that owner-managers provide more advice to entrepreneurs than captive managers (such as bank's venture capital divisions).

However, hypothesis H7b is infirmed: Funds with long-term investors, in particular, pension-funds, as Limited Partners do not appear more likely to implement responsible practices. Table 3 displays that firms with Pension funds as Limited Partners, as well as Institutional and Sovereign Limited Partners, are not more likely to be UNPRI signatories.

However, ownership seems to matter more than Limited Partners in this regard. CSR Communication is also more impacted by who owns the Private Equity firm rather than who finances it (Table 6).

Results on Responsive Drivers of Socially Responsible Private Equity

Results partly confirm hypothesis H8 on responsive driver of socially responsible investing: Large and thus visible

Private Equity firms are more likely to engage in socially responsible practices that are well formalized and publicly communicated to protect their reputation and license-to-operate; yet, LBO specialists do not particularly stand out in this regard.

We previously noted that large Private Equity firms in terms of workforce were more likely to be UNPRI signatories, and this effect appears stronger and more robust when size is captured by the amounts of assets managed. Indeed, marginal effects observed indicate that a 1 unit increase in the logarithm of assets managed increases the probability of being a UNPRI signatory by 1.98 % (model 4, Table 4, $p < 0.01$; effect of 1.84 % in model 3). In other words, a 100 % increase in assets managed increases the probability of being a UNPRI signatory by 1.98 %. Results also show that a 100 % increase in assets managed increases the probability of public communication on responsible investing by 3.53 % (model 3, Table 6, $p < 0.01$; effect of 2.65 % in model 4). Let us note that despite significance, the size effect magnitude appears small (in particular, compared to the independence effect previously discussed).

Survey data also show that ESG Chart, ESG dedicated post, ESG reporting, and UNPRI signature come as a bundle (see correlation Table 8 in the Appendix). Hereby, formalizing socially responsible investing appears more likely in large Private Equity firms.

However, being specialized on Transmission capital (hence LBO) does not appear to have any significant effect on UNPRI signature and communication.

In our results, geographical scope variables appear to play a significant role negatively for PRI signature (at the international scope), and positively for CSR communication (at the European scope), suggesting that various national contexts may affect SRI behaviors. In particular, the positive effect of investing at the European scope on CSR communication may suggest a form of mainstreaming of SRI at the European level (for instance, influenced by the CSR reporting constraints imposed in various countries); whereas the negative effect of investing at an

international scope on PRI signature may suggest that the diversity of institutional contexts does not favor transversal and global SRI engagement.

How to disentangle differentiation from reputation protection? First, let us note that 46 % survey respondents believe ESG matter for reputation and image risk (59 % UNPRI signatories), which is high but less than the belief in ESG importance for Limited Partners' demand. Second, we observe that very few firms communicate their ESG reporting (1 %) and ESG Chart (5 %) to the public. Those elements lead us to conjecture that protecting firm's image and reputation matters more for the business than for the public.

Limits of the Analysis and Further Research Paths

This exploratory article presents evidence on SRI at the Private Equity firm level and shows an important role for engagement.

Several other factors, which were beyond the scope of our database, are likely to also influence SRI behavior in Private Equity. Compensation terms, for instance, might influence socially responsible practices. Indeed, a related literature has shown the role of executive compensation in the implementation of CSR in listed companies (e.g., Mahoney and Thorne 2006; Berrone and Gomez-Mejia 2009). Besides, Johan and Najar (2010) examine the role of law and culture in setting fund manager fees and show that they are much more significant than manager characteristics and/or market conditions. Interactions between legal conditions, culture, compensation terms, and engagement would thus be highly interesting to investigate in further research.

Similarly, we control for education and international experience of the fund manager, but not experience. Whether the fund manager is experienced or a first time fund manager might impact his investment practices. Though fund manager education and international experience do not appear to play a significant role in our approach, experience may affect responsible investment practices (see Cumming and Walz 2010).

SRI behavior in Private Equity could also be driven by other factors, such as regulatory pressure. An extensive literature has highlighted that the threat of fines and other regulatory costs may induce higher CSR activities (for instance, see Lutz et al. 2000). Socially responsible practices might thus also have been influenced by regulatory pressure. In particular, in France, the recently published Grenelle II Law (April 2012) now requests non-listed firms to publish ESG reporting (starting, respectively, from 2012 to 2014 for firms employing above 5000 and 500 employees). Extending our research to other markets, such as in the United Kingdom or the United States, would provide an effective test of the importance of the regulatory and institutional context on Private Equity socially

responsible investing. Moreover, it would test for the national anchorage of our results.

From a wider perspective, our study highlights important strategic drivers behind socially responsible Private Equity. Hereby, ESG concerns tackled by investors are likely to be those identified as having the largest impact on the bottom line of the companies they owned. Testing this hypothesis would constitute a promising extension of our research. Such an analysis could be lead at the company level, for instance, by exploring the impact investor engagement has on the company, its stakeholders and society as a whole.

Finally, another level that could be worth investigating would be the Private Equity fund level. Within the same Private Equity firm, it is indeed likely that differences exist between funds managed. Funds can for instance vary depending on their vintage year, as it is related to market conditions and to business cycles and is known to impact company valuations and return expectations (see Kaplan and Schoar 2005). Funds can also undergo style drift, meaning they can deviate from their stated objectives in terms of the focus for their investments. Cumming et al. (2009) have shown that style drifts are less likely among younger fund managers, are affected by market conditions, and are positively related to investment performance. Focusing on the fund level would enable to identify whether responsible practices are impacted by style drift, and conversely.

Discussion

We provide a summary of our hypotheses and results in Table 7 and discuss below our main observations from this empirical analysis.

Socially Responsible Private Equity: Responsive or Strategic?

Our empirical results support that responsible investment in French Private Equity is a mainstream movement which got quickly structured under the impetus of independent large conventional actors, both in terms of human capital and assets managed.

This movement in fact appears characterized by a mix of both ESG screening and engagement, with Private Equity investors typically being much involved in the portfolio company's management.

Socially responsible investing seems essentially strategically driven, Private Equity investors hereby aiming at creating value, improving risk management and differentiating to raise fund, especially independent firms. Among all the effects we tested, we found that firm independence was by far the largest in terms of magnitude. SRI as

Table 7 Summary of hypotheses and results

Hypotheses	Test result	Conclusion (if hypothesis is not confirmed)
H1: Socially responsible PE firms tend to integrate ESG issues in mainstream business, rather than considering it as a niche market segment.	Confirmed	
H2a: PE firms are more likely to favor Governance aspects within the ESG issues.	Confirmed	
H2b: Socially responsible PE heavily relies on investor engagement.	Confirmed	
H3: The integration of ESG issues is more likely to be implemented in large PE firms in terms of workforce.	Partly confirmed	The integration of ESG issues is more likely to be implemented in large PE firms in terms of workforce, yet the effect is small.
H4: Socially responsible investing in PE is motivated by creating value in the companies owned.	Partly confirmed	Socially responsible investing in PE are motivated by creating value in the companies owned, yet not only (for 43 % of the respondents).
H5: Socially responsible investing in PE is a risk-management tool.	Confirmed	
H6: Venture capitalists are more likely than other PE investors to develop specialized green funds to open new markets.	Confirmed	
H7a: Independent funds are more likely than captive funds to integrate ESG issues as a differentiation tool to attract investors.	Confirmed	
H7b: Funds with long-term investors, in particular, pension-funds, as Limited Partners are more likely to integrate ESG issues.	Informed	Funds with long-term investors, in particular pension-funds, as Limited Partners do not appear more likely to implement responsible practices.
H8: Large and thus visible PE firms as well as LBO specialists are more likely to formalize socially responsible investing and to publicly communicated it to protect their reputation and license-to-operate.	Partly confirmed	Large and thus visible PE firms are more likely to engage in socially responsible practices that are well formalized and publicly communicated to protect their reputation and license-to-operate; yet LBO specialists do not particularly stand out in this regard.

This table summarizes our hypotheses, their test results and our conclusions when the hypotheses were only partly confirmed or informed

implemented by those investors, therefore, appears in line with shareholders' value maximization and consistent with a business improvement.

Socially Responsible Private Equity: Engagement or Activism?

Our approach confirms that engagement on ESG issues quite differs between Private Equity investors and Public investors. On public markets, engagement is usually referred to as shareholder activism. Sparkes (2008) defines shareholder activism as "the use of the voting rights attached to ordinary shares to influence company management." As they usually are majority or significant shareholders, Private Equity investors have considerable influence as active investors.

Moreover, engagement in the specific context of Private Equity is to be distinguished from conventional ethical activism, such as undertaken by religious institutional investors in the United States through the Interfaith Center on Corporate Responsibility (Williams 2007). According to Smith (1996), engagement acts as an implicit threat to the company that if the investor's concerns are not addressed, an exclusionary strategy may be adopted. Sparkes (1998),

hence, have argued that shareholder activism differs from SRI as advocacy is characterized by a single-issue focus, no financial concerns, and the seeking of confrontation and publicity; whereas SRI is characterized by multi-issue concern, strong financial interest, the seeking of engagement and the avoidance of publicity. However, Private Equity investors are active investors who promote ESG issues on strategic grounds rather than ethical ones. Hereby, their engagement does not fit to Sparkes's definition as they target promoting CSR to maximize shareholders' value. A possible explanation of this difference might lie in the specificity of the French context, whereby SRI mainstreaming has been taking place under the explicit goal of penetration of ESG criteria into conventional funds, thereby maybe creating a sort of 'ESG externality' on the socially Responsible Private Equity market.

Socially Responsible Private Equity: A Stronger Impact than Public Investors

Finally, this article's findings on Private Equity investors suggest that those who care for ESG issues might have a stronger impact as active investors than Public investors. Indeed, results are that 23 % of respondents brought ESG

issues to company's supervisory board, 53 % directly monitored those issues, and 26 % visited plants. Compared to such figures, shareholder advocacy is still in practice limited on public markets. For instance, Lewis and Mackenzie (2000) found little support of hard engagement in their large survey of UK ethical public investors, passive signaling (screening) being most favored. In the United States, Lydenberg (2002) argues that the "Wall Street Rule" generally applies, that is shareholders are expected to regulate managers by selling shares rather than by trying to change management practices through engagement. Indeed, SRI funds which used both screening and advocacy accounted in 2003 for about a fifth of all SRI funds (Social Investment Forum 2003). In Europe, Eurosif (2010 European SRI Survey) estimated € 1.5 trillion assets managed were impacted by SRI engagement, representing about 28 % of all broad SRI approaches in Europe.

Here we find that conventional Private Equity funds which care for ESG issues appear active at an equivalent level on the French market. Hereby findings hint that this financial industry might have the potential to become a powerful tool to promote sustainable practices in portfolio companies. However, to realize this potentiality, Private Equity firms need to develop a sound expertise of ESG issues and to acquire the corresponding human capital. Considering the current difficulties of the industry related to the financial crisis and liquidity dearth, it seems unlikely that Private Equity firms will currently invest in such competencies. Would that turn out so, the development of an effective and reliable responsible investment movement beyond its current frontiers might be compromised.

Conclusion

Whereas most SRI literature has focused on public markets, this article highlights that French Private Equity investors recently seized ESG issues and are developing a mainstream responsible investing approach structurally based on engagement. As significant shareholders, Private Equity investors have the potential to actively promote sustainable practices in the firms they own. Findings support that such activism is strategically grounded, as managing portfolio ESG issues might enhance value creation, enlarge risk management and enable Private Equity firms to differentiate to raise funds. Considering the specificities of the studied market, a promising research path would be to explore whether this responsible investing movement is restricted to France or whether drivers and maturation differ between Private Equity markets.

At a time of financial crisis and regulation stringency for Private Equity, responsible investment hence appears to have been "thought global" by large leading actors to improve the

mainstream business and provide it with new growth tools. However, most Private Equity companies currently seem to lack the human capital and expertise essential for successfully implementing a profitable ESG issues management. With financial markets still in the crisis turmoil and the upcoming of tougher regulatory standard on bank capital adequacy (Basel III), the European Private Equity industry environment will quickly evolve in the next few months. The question of how the new Private Equity responsible investing movement will react to this shifting context, and potential consequences for firm managers' access to capital, henceforth still lay wide open.

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Appendix: Survey Wording

"ESG Practices of Private Equity Investors—Survey 2011" (Novethic)

ESG Policy at the Management Firm Level

1. Have you formalized a policy or a chart stating how you take into account Environmental, Social and Governance (ESG) criteria in your portfolio and/or your investments?
Yes; Please detail since when/No, but we plan to do so this year (directly go to question 5)/No (directly go to question 5)
2. If you answered yes to question 1, to whom do you communicate it?
Our co-workers/Our investors/Our portfolio companies/Publicly communicated
3. Are you a signatory of the French Private Equity Association Chart?
Yes/No; why?
4. Are you a signatory of the United Nations Principles for Responsible Investments (PRI)?
Yes; Please detail since when/No, but we plan to do so this year/No.
5. Are your co-workers trained in ESG issues management in their daily business?
Yes; Please detail how/No.
6. Your ESG policy is run by:
A co-worker dedicated to this policy/A co-worker non-dedicated to this policy/Investment directors/An external third-party/Other, please detail.
7. Within your management firm, which policies have you launched in terms of sustainable development?

ESG evaluation at the management firm level

Environment: Management firm carbon footprint evaluation/Videoconferences/Energy savings thanks to eco-gestures

Social: employee training/annual job appraisal/employee profit-sharing

Governance: Ethics Chart or Deontology Code

Other: please detail

Implementation in Portfolio Companies

8. Do you exclude from your investments:
Specific industries: Weapons/Alcohol/Tobacco/Pornography/Hazard games/GMOs/Nuclear Energy/Other, please detail
Specific policies considered reprehensible: Child labor/Forced labor/Other, please specify
9. Which means do you use to evaluate ESG practices in companies (before investment; over the investment phase; while the company is hold in your portfolio)?
 Dialogue between the management firm and the portfolio company/ESG survey sent to the company/Management firm internal ESG audit grid/Acquisition dues diligences/Specific acquisition dues diligences/Audit by an external third-party/Other, please detail.
10. In this context, are environmental issues (water and energy consumptions, carbon assessment, waste management, etc.) evaluated:

	Always	Sometimes *	Never	No, but we plan to do so this year*
When you acquire a company				
When you hold a company in portfolio				
When you sell a company				

*Please detail how many companies were evaluated in 2010.

1. In this context, are social issues (training, skill management, diversity, non discrimination, etc.) evaluated:

	Always	Sometimes *	Never	No, but we plan to do so this year*
When you acquire a company				
When you hold a company in portfolio				
When you sell a company				

*Please detail how many companies were evaluated in 2010.

1. In this context, are governance issues (allocation and delegation of authority, board formation, board independence, fight against corruption, etc.) evaluated:

	Always	Sometimes *	Never	No, but we plan to do so this year*
When you acquire a company				
When you hold a company in portfolio				
When you sell a company				

*Please detail how many companies were evaluated in 2010.

2. How do you support your portfolio companies to implement their sustainable development policy?
 Regular dialogue between the management firm and a dedicated contact in the portfolio company/Link establishment with a specialized consultant/Support of the ESG reporting/Follow-up, but the portfolio company has full autonomy to implement the policy/Other, please detail.
3. How do you ensure ESG criteria are taken into account in the companies in which you invest?
 Chart compliance request/ESG reporting request/ESG issues on the agenda of the company's supervisory board/Visit of plants/ESG clause in the shareholders pact/Other; please detail.
4. Have you ever measured the impact of the ESG issue management on your portfolio?
 Yes; please detail/No.
5. The ESG evaluation of a company already led you to:
 Decrease a target company valuation/Reject an investment/The company ESG evaluation had no impact on our management/Other; please detail.

Relationships with Investors

6. Have you implemented a regular ESG reporting of your portfolio?
 Yes, it is public/Yes, it is communicated to our investors/No/No, but we plan to do so this year.
7. Do your Limited Partners ask you how you manage ESG issues?
 Yes, often/Yes, sometimes/No, never.
8. If you answered yes to question 7, how have your Limited Partners formalized their request (survey, side letters, shareholders pact, etc.)?

Motivations

9. For which reasons do you take into account ESG issues in your investments?
 To improve risk management (social litigations, environmental liability, etc.)/To answer our investors' demand/To ease fund raising/To improve deal flow/To reduce the management firm reputational risk/to improve the value of portfolio companies/other, please detail.

See Tables 8 and 9.

Table 8 Correlation matrix (1/3)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1. Firm Age	1.00													
2. Workforce	0.15	1.00												
3. Funds	0.02	0.45	1.00											
4. Companies	0.16	0.39	0.70	1.00										
5. Assets managed	0.14	0.39	0.26	0.32	1.00									
6. Listed	0.21	0.04	-0.08	-0.06	0.08	1.00								
7. PE firm	0.00	0.03	-0.01	-0.08	-0.03	-0.03	1.00							
8. Partners	-0.14	-0.13	-0.01	-0.23	0.08	-0.10	-0.38	1.00						
9. Bank	0.06	0.10	0.12	0.30	-0.08	-0.08	-0.23	-0.43	1.00					
10. Insurance	0.04	0.12	0.10	0.27	0.18	-0.03	-0.09	-0.18	-0.06	1.00				
11. Industry	0.01	-0.06	-0.07	-0.04	-0.06	-0.06	-0.12	-0.22	-0.08	-0.04	1.00			
12. French state	-0.01	0.01	-0.07	0.06	-0.09	-0.04	-0.15	-0.37	-0.03	-0.02	-0.03	1.00		
13. Other	0.07	-0.04	-0.10	-0.07	-0.02	0.06	-0.09	-0.16	-0.08	-0.01	-0.04	-0.00	1.00	
14. LP Industry	-0.05	-0.08	-0.09	-0.04	-0.13	-0.08	-0.05	-0.12	-0.10	-0.07	0.51	0.16	-0.08	1.00
15. LP Captive	0.00	0.03	0.10	0.17	-0.00	0.06	0.13	-0.32	0.48	0.14	-0.14	-0.22	-0.04	-0.19
16. LP Sovereign	-0.07	0.02	-0.02	0.11	-0.12	-0.06	-0.08	-0.19	-0.01	0.02	-0.08	0.55	-0.02	0.29
17. LP Individuals	-0.04	0.20	0.50	0.26	0.13	0.05	-0.01	0.11	0.01	-0.05	-0.05	-0.19	0.06	-0.11
18. LP Institutional	-0.03	0.02	0.01	-0.09	0.03	-0.02	-0.03	0.30	-0.24	-0.07	-0.13	-0.01	-0.04	-0.01
19. LP Pension funds	0.08	0.22	0.03	0.02	0.30	0.08	0.10	0.20	-0.19	0.04	-0.13	-0.14	-0.04	-0.05
20. LP Family offices	0.03	-0.02	-0.07	-0.10	-0.10	0.05	-0.10	0.19	-0.18	-0.02	-0.07	0.00	0.07	0.13
21. VC	-0.02	0.05	0.26	0.16	-0.09	-0.05	0.01	-0.14	-0.01	-0.10	0.06	0.28	0.04	0.20
22. Transmission	0.17	0.13	-0.01	0.07	0.15	0.08	-0.05	0.02	0.01	0.05	0.01	-0.06	0.03	-0.04
23. Growth	0.04	0.08	0.13	0.17	-0.11	-0.05	-0.01	-0.09	0.19	-0.03	-0.02	-0.03	-0.01	0.05
24. Mezzanine	0.02	0.22	0.02	0.10	0.12	0.04	0.06	-0.08	-0.05	0.14	0.09	-0.05	0.00	-0.02
25. Distressed Capital	-0.04	0.15	-0.08	-0.12	0.02	-0.02	0.07	0.10	-0.09	-0.05	-0.08	-0.06	-0.03	-0.09
26. Funds of funds	-0.10	0.25	0.27	0.16	0.01	-0.05	0.14	-0.08	0.04	0.15	-0.09	-0.06	-0.08	-0.13
27. Minority	0.05	0.05	0.16	0.24	-0.26	-0.06	0.05	-0.25	0.21	0.12	-0.03	0.15	-0.07	0.09
28. Regional	0.06	-0.08	-0.10	0.09	-0.13	-0.07	-0.09	-0.35	0.24	0.01	0.03	0.44	-0.02	0.18
29. European	-0.03	0.15	0.17	-0.01	0.22	0.11	0.04	0.23	-0.18	0.03	-0.03	-0.23	-0.03	-0.11
30. International	0.01	0.23	0.16	0.12	0.22	0.21	0.03	0.03	-0.04	0.08	-0.01	-0.12	-0.07	-0.03
31. Sector	-0.08	-0.08	0.03	-0.14	-0.04	-0.07	0.06	0.07	-0.13	-0.07	0.11	0.01	-0.06	0.12
32. Gender	-0.08	0.00	0.02	0.09	0.11	0.09	-0.10	-0.08	-0.04	0.09	0.14	0.12	0.02	0.04
33. Founder	-0.20	-0.02	-0.06	-0.13	-0.10	-0.10	-0.12	0.52	-0.33	-0.03	-0.08	-0.27	-0.02	-0.05
34. Engineer	0.03	-0.02	0.03	0.09	0.11	-0.03	-0.04	-0.01	0.02	0.05	-0.08	0.14	0.00	-0.08
35. Business school	-0.06	-0.10	0.03	-0.02	-0.13	0.09	-0.01	0.01	0.02	-0.03	0.18	-0.17	-0.08	0.12

Table 8 continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
36. mgt International	-0.09	0.12	-0.01	-0.10	0.04	0.04	0.09	0.07	-0.10	-0.09	0.03	-0.13	-0.05	-0.08
37. mgt Other	0.03	-0.06	-0.03	0.01	-0.01	-0.07	-0.04	-0.02	0.08	-0.04	-0.06	0.07	0.06	0.07
38. UNPRI	0.06	0.15	0.19	0.14	0.20	-0.01	0.04	0.07	-0.08	0.01	-0.03	-0.05	-0.07	-0.04
39. AFIC Chart	-0.04	-0.06	0.13	0.05	0.02	0.05	-0.00	0.24	-0.06	0.04	-0.03	-0.36	-0.02	-0.10
40. Green or Social fund	-0.07	-0.03	0.05	-0.03	-0.06	-0.06	-0.04	-0.05	-0.05	0.01	0.12	0.05	0.13	0.07
41. Communication	0.01	0.20	0.16	0.07	0.27	0.06	-0.06	0.07	-0.12	0.09	-0.01	-0.06	0.14	0.00
42. Interest	0.04	0.11	0.18	0.12	0.11	-0.02	-0.01	-0.02	-0.07	0.02	0.06	0.04	0.09	0.06
15. LP Captive	1.00													
16. LP Sovereign	-0.17	1.00												
17. LP Individuals	-0.01	-0.04	1.00											
18. LP Institutional	-0.27	0.11	0.01	1.00										
19. LP Pension funds	-0.20	-0.04	-0.03	0.17	1.00									
20. LP Family offices	-0.25	0.20	-0.04	0.13	0.10	1.00								
21. VC	-0.09	0.35	0.23	-0.06	-0.14	-0.02	1.00							
22. Transmission	-0.08	-0.12	0.00	0.11	0.19	0.01	-0.24	1.00						
23. Growth	0.05	0.11	0.16	-0.06	-0.11	-0.08	-0.01	0.28	1.00					
24. Mezzanine	0.12	-0.08	-0.08	-0.02	-0.06	0.03	-0.16	-0.01	-0.03	1.00				
25. Distressed Capital	-0.06	-0.04	-0.07	0.06	0.08	-0.04	-0.12	0.11	-0.01	-0.04	1.00			
26. Funds of funds	0.11	-0.14	0.01	-0.09	0.08	-0.08	-0.09	-0.15	-0.14	0.07	0.01	1.00		
27. Minority	0.10	0.33	0.15	-0.14	-0.14	-0.01	0.36	-0.28	0.33	0.07	-0.26	0.12	1.00	
28. Regional	-0.01	0.35	-0.02	-0.01	-0.25	-0.04	0.18	0.10	0.18	-0.09	-0.08	-0.13	0.26	1.00
29. European	-0.06	-0.23	-0.05	0.04	0.31	0.02	-0.10	-0.11	-0.16	0.08	0.02	0.23	-0.20	-0.47
30. International	0.03	-0.09	0.01	0.09	0.18	-0.05	0.09	-0.06	0.05	0.08	0.05	0.21	0.07	-0.23
31. Sector	-0.07	0.03	-0.03	0.10	0.04	-0.03	0.24	-0.29	-0.08	-0.07	-0.06	-0.03	-0.03	-0.18
32. Gender	0.01	0.12	-0.03	-0.05	-0.04	-0.05	0.04	-0.02	-0.05	0.06	-0.07	0.04	0.06	-0.02
33. Founder	-0.22	-0.08	0.12	0.20	0.07	0.08	-0.04	-0.06	-0.03	0.01	0.10	-0.07	-0.05	-0.24
34. Engineer	0.04	0.12	0.00	-0.04	0.07	-0.03	-0.00	-0.08	-0.02	0.01	-0.04	-0.03	-0.03	-0.14
35. Business school	0.13	-0.02	0.15	-0.11	-0.08	0.15	-0.02	-0.07	0.05	0.05	-0.11	0.03	0.05	-0.07
36. mgt International	-0.09	-0.13	-0.12	0.13	0.11	-0.13	-0.09	0.10	-0.05	0.02	0.15	-0.02	-0.18	-0.10
37. mgt Other	-0.10	0.03	0.02	0.08	-0.08	0.04	0.11	-0.10	-0.04	-0.12	-0.07	-0.04	0.12	0.15
38. UNPRI	0.06	-0.06	0.07	0.06	0.17	-0.08	-0.06	0.03	-0.04	-0.00	-0.10	0.08	-0.02	-0.11
39. AFIC Chart	0.18	-0.22	0.16	-0.01	0.04	-0.04	-0.12	-0.01	0.03	0.03	0.01	0.02	-0.15	-0.30
40. Green or Social fund	0.05	0.07	0.05	-0.07	-0.11	-0.11	0.17	-0.18	0.13	-0.10	-0.00	-0.02	0.07	0.02

Table 8 continued

	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)
41. Communication	0.05	-0.06	0.12	0.00	0.13	-0.15	0.02	-0.01	0.02	-0.06	-0.11	0.03	-0.11	-0.14
42. Interest	0.04	0.03	0.02	0.07	0.07	0.01	-0.01	0.10	0.01	0.04	-0.01	-0.02	-0.10	-0.03
29. European	1.00													
30. International	0.37	1.00												
31. Sector	0.18	0.07	1.00											
32. Gender	0.03	0.01	-0.09	1.00										
33. Founder	0.08	-0.01	0.13	0.01	1.00									
34. Engineer	0.06	-0.00	0.15	-0.10	-0.12	1.00								
35. Business school	0.04	-0.05	-0.00	0.09	-0.07	-0.24	1.00							
36. mgt International	0.12	0.15	-0.00	0.06	0.10	-0.13	-0.21	1.00						
37. mgt Other	-0.10	0.07	0.05	-0.02	-0.04	-0.13	-0.28	-0.31	1.00					
38. UNPRI	0.14	-0.02	0.03	0.04	0.01	0.04	-0.10	-0.00	0.02	1.00				
39. AFIC Chart	0.22	0.15	-0.01	-0.08	0.13	0.01	0.13	-0.00	-0.07	0.15	1.00			
40. Green or Social fund	-0.02	0.09	0.34	0.02	-0.01	0.05	-0.01	0.01	0.17	0.03	0.04	1.00		
41. Communication	0.17	0.16	0.13	0.02	0.07	0.13	-0.13	0.03	0.07	0.37	0.13	0.42	1.00	
42. Interest	0.04	-0.06	0.02	-0.02	-0.04	0.02	0.04	-0.17	0.05	0.40	0.11	0.15	0.30	1.00

This table presents the Pearson's correlation coefficients between variables of the public data database. Correlations significant at the 5 % level or lower are underlined

Table 9 Survey sample representativeness

Variable name	Definition	Survey respondents			Survey non-respondents			Comparison of tests
		Obs.	Mean	Median	Obs.	Mean	Median	
Main characteristics								
Firm age	Years elapsed since firm foundation	74	13.9	11.0	227	12.9	11.0	1.0
Workforce	Number of employees	68	18.6	10.5	218	11.5	6.0	7.1**
Funds	Number of funds managed by the firm	74	7.7	2.5	228	4.1	2.0	3.7***
Companies	Number of companies hold in portfolio	68	55.7	17.5	220	34.8	16.0	20.9**
Assets managed	Millions of Euros of assets managed	72	1422	318	223	700	100	722**
Ownership: percentage of firms shares...								
Listed	... listed on a Stock market (in %)	73	1.1	0.0	227	1.6	0.0	−0.0
PE firm	... owned by another Private Equity firm (in %)	70	15.5	0.0	204	16.1	0.0	−0.0
Partners	... owned by the firm Partners (in %)	70	40.7	22.5	201	42.3	0.0	−0.0
Bank	... owned by a Bank (in %)	70	13.9	0.0	201	18.8	0.0	−0.1
Insurance company	... owned by an insurance company (in %)	70	3.8	0.0	201	3.2	0.0	0.0
Industry	... owned by an industrial corporation (in %)	70	7.3	0.0	201	4.7	0.0	0.0
French state	... owned directly or indirectly by the French State (in %)	71	13.4	0.0	203	10.8	0.0	0.0
Other	... owned by other shareholders (in %)	70	5.5	0.0	201	2.7	0.0	0.0
Limited Partners: dummy variable equal to 1 if..., 0 otherwise								
LP Industry	...the firm manages funds provided by an industrial corporation	69	0.23	0.00	199	0.18	0.00	0.06
LP Captive	... the firm is captive and essentially manages funds provided by a single bank or insurance	69	0.35	0.00	199	0.31	0.00	0.04
LP Sovereign	...the firm manages funds provided by a Sovereign Wealth Funds	71	0.41	0.00	200	0.38	0.00	0.03
LP Individuals	... the firm manages funds provided by individuals	71	0.23	0.00	199	0.21	0.00	0.01
LP Institutional	... the firm manages institutional funds (except pension funds)	70	0.87	1.00	194	0.76	1.00	0.11**
LP Pension funds	... the firm manages funds provided by pension funds	70	0.31	0.00	194	0.25	0.00	0.07
LP Family offices	... the firm manages funds provided by family offices	69	0.38	0.00	194	0.37	0.00	0.01
Activity: dummy variable equal to 1 if..., 0 otherwise								
Venture Capital	... funds are invested in venture capital	74	0.42	0.00	226	0.42	0.00	−0.01
Transmission	...funds are invested in transmission capital	74	0.69	1.00	226	0.57	1.00	0.12**
Growth	...funds are invested in growth capital	74	0.61	1.00	226	0.59	1.00	0.02
Mezzanine	...funds are invested in mezzanine	74	0.09	0.00	226	0.07	0.00	0.03
Distressed Capital	...funds are invested in distressed capital	74	0.08	0.00	226	0.09	0.00	−0.00
Funds of funds	...funds are invested in funds of funds	74	0.09	0.00	226	0.11	0.00	−0.02
Investment scope: dummy variable equal to 1 if..., 0 otherwise								
Minority	... the firm invests as a minority shareholder	65	0.71	1.00	174	0.80	1.00	−0.10*
French district scope	... the firm invests only in a specific French district	74	0.15	0.00	234	0.18	0.00	−0.03
European scope	... the firm invests in other European countries than France	74	0.61	1.00	234	0.57	1.00	0.04
International scope	... the firm invests in other countries than European countries	74	0.16	0.00	234	0.22	0.00	−0.06
Sector	... the firm manages at least one fund specialized in a sector	72	0.38	0.00	227	0.36	0.00	0.02

Table 9 continued

Variable name	Definition	Survey respondents			Survey non-respondents			Comparison of tests
		Obs.	Mean	Median	Obs.	Mean	Median	
Management: dummy variable equal to 1 if the Chairman..., 0 otherwise								
Gender	... is a woman	74	0.04	0.00	225	0.05	0.00	−0.00
Founder	... founded the firm	73	0.40	0.00	217	0.44	0.00	−0.04
Engineer	... has a French “Grandes Ecoles” Engineer background	71	0.30	0.00	185	0.28	0.00	0.02
Business school	...has a French Business School background	71	0.51	1.00	183	0.46	0.00	0.05
International	... has an international background	71	0.11	0.00	183	0.27	0.00	−0.16***
Other	... has a different French background	71	0.49	0.00	183	0.44	0.00	0.05
Public responsible investment practices: dummy variable equal to 1 if the firm..., 0 otherwise								
UNPRI	... is a UNPRI signatory	74	0.30	0.00	234	0.03	0.00	0.27***
AFIC Chart	... is signatory of the French Private Equity Association Ethics Chart	74	0.77	1.00	234	0.65	1.00	0.12**
Green or Social fund	... manages at least one fund with a social or environmental target	72	0.21	0.00	227	0.09	0.00	0.12***
Communication	...has a website referring to responsible investing practices	74	0.38	0.00	223	0.11	0.00	0.27***

This table compares the main characteristics of survey respondents and survey non-respondents based on the public data we gathered (see Table 1). We test for survey sample representativeness of the French Private Equity industry by using *t* tests for normally distributed variables and tests of proportion for dummy variables: respondents significantly differ from non-respondents at the *** 1 %, ** 5 %, * 10 % levels

References

- Amess, K., & Wright, M. (2007). The wage and employment effects of leveraged buyouts in the UK. *International Journal of the Economics of Business, Special Issue on Control Transactions: Productivity, Wage and Employment Implications*, 14(2), 179–195.
- Arjaliès, D. A. (2010). A social movement perspective on finance: How socially responsible investment mattered. *Journal of Business Ethics*, 92(1), 57–78.
- Arora, S., & Gangopadhyay, S. (1995). Toward a theoretical model of voluntary overcompliance. *Journal of Economic Behavior & Organization*, 28(3), 289–309.
- Association Française des Investisseurs en Capital (AFIC), & Thornton, Grant. (2010). *L'Activité du capital investissement Français*. Paris: AFIC.
- Association Française des Investisseurs en Capital (AFIC), & Thornton, Grant. (2011). *Etude premier semestre 2011. L'Activité du capital investissement Français*. Paris: AFIC.
- Association for Sustainable and Responsible Investment in Asia (ASRIA). (2011). *Sustainable Investment in Asian emerging markets*. Washington, DC: International Finance Corporation.
- Baron, D. P., Harjoto, M. A., & Jo, H. (2011). The economics and politics of corporate social performance. *Business and Politics*, 13(2), 1–46.
- Bauer, R., Koedijk, K., & Otten, R. (2005). International evidence on ethical mutual fund performance and investment style. *Journal of Banking & Finance*, 29(7), 1751–1767.
- Berrone, P., & Gomez-Mejia, L. R. (2009). The pros and cons of rewarding social responsibility at the top. *Human Resources Management*, 48(6), 959–971.
- Bocquet, A. (2007). *Rapport sur la proposition de résolution (no 3491) de M. Alain Bocquet, Tendant à la Création d'une Commission d'Enquête sur la Place et le Rôle des Fonds d'Investissement dans l'Economie, sur leurs Méthodes d'Acquisition d'Entreprises par Effet de Levier appelées LBO, sur les Conséquences de Telles Pratiques pour l'Emploi, les Salaires et les Conditions de Travail ainsi que sur les Solutions Alternatives qui Pourraient Être Mises en Œuvre à Partir de la Constitution d'un Pôle Financier Public*. Paris: Assemblée Nationale.
- Bollen, N. P. B. (2007). Mutual fund attributes and investor behaviour. *Journal of Financial and Quantitative Analysis*, 42, 689–708.
- Boucly, Q., Sraer, D., & Thesmar, D. (2009). Job Creating LBOs'. Working Paper, Berkeley University and HEC Paris. Available at SSRN: <http://ssrn.com/abstract=1354087>.
- Bourghelle, D., Hager, J., & Louche, C. (2009). The integration of ESG information into investment processes: Toward an emerging collective belief? Working Paper, Vlerick Leuven Gent Management School. <http://ideas.repec.org/p/vlg/vlgwps/2009-26.html>.
- Brammer, S., Brooks, S., & Pavelin, S. (2006). Corporate social performance and stock returns: UK evidence from disaggregate measures. *Financial Management*, 35(3), 97–116.
- Brau, J. C., & Fawcett, S. E. (2006). Initial public offerings: An analysis of theory and practice. *Journal of Finance*, 61(1), 399–436.
- Brekke, K. A., & Nyborg, K. (2008). Attracting responsible employees: Green production as labor market screening. *Resource and Energy Economics*, 30(4), 509–526.
- British Venture Capital Association (BVCA), & Ipsos MORI. (2009). *BVCA: Members views of sustainability issues*. London: BVCA. http://admin.bvca.co.uk/library/documents/Respondent_report.pdf.
- British Venture Capital Association (BVCA), PriceWaterhouse Coopers, & Waterman. (2010). *Responsible investment. A guide for private equity and venture capital firms*. London: BVCA.
- Cespa, G., & Cestone, G. (2007). Corporate social responsibility and managerial entrenchment. *Journal of Economics & Management*

- Strategy, Volume Corporate Social Responsibility and Managerial Entrenchment, 16(3), 741–771.
- Climent, F., & Soriano, P. (2011). Green and good? The investment performance of US environmental mutual funds. *Journal of Business Ethics*, 103(2), 275–287.
- Cox, P., Brammer, S., & Millington, A. (2004). An empirical examination of institutional investor preferences for corporate social performance. *Journal of Business Ethics*, 52(1), 27–43.
- Cumming, D. J., Fleming, G., & Schwienbacher, A. (2008). Financial intermediaries, ownership structure and the provision of venture capital to SMEs: Evidence from Japan. *Small Business Economics*, 31(1), 59–92.
- Cumming, D. J., Fleming, G., & Schwienbacher, A. (2009). Style drift in private equity. *Journal of Business Finance and Accounting*, 36(5–6), 645–678.
- Cumming, D. J., & Johan, S. (2007). Socially responsible institutional investment in Private Equity. *Journal of Business Ethics*, 75(4), 395–416.
- Cumming, D. J., Johan, S. A., Fleming, G., & Takeuchi, M. (2010). Legal protection, corruption, and Private Equity returns in Asia. *Journal of Business Ethics*, 95(2), 173–193.
- Cumming, D. J., & Walz, U. (2010). Private equity returns and disclosure around the world. *Journal of International Business Studies*, 41, 727–754.
- Cumming, D. J., & Zambelli, S. (2010). Illegal buyouts. *Journal of Banking & Finance*, 34(2010), 441–456.
- Cumming, D. J., & Zambelli, S. (2012). Private equity performance under extreme regulation. *Journal of Banking and Finance*. doi: 10.1016/j.jbankfin.2012.04.002.
- Davis, S., Haltiwanger, J., Jarmin, R. S., Lerner, J., & Miranda, J. (2011). Private Equity and employment. NBER Working Papers No 17399, National Bureau of Economic Research. <http://EconPapers.repec.org/RePEc:nbr:nberwo:17399>.
- Derwall, J., Gunster, N., Bauer, R., & Koedijk, K. (2005). The eco-efficiency premium puzzle. *Financial Analysts Journal*, 61(2), 51–63.
- Desbrières, P., & Schatt, A. (2002). The impacts of LBOs on the performance of acquired firms: The French Case. *Journal of Business Finance and Accounting*, 29(5–6), 695–729.
- Dunfee, T. W. (2003). A critical perspective of integrative social contracts theory: Recurring Criticisms and next generation research topics. *Journal of Business Ethics*, 68(3), 303–328.
- Eccles, N. S., & Viviers, S. (2011). The origins and meanings of names describing investment practices that integrate a consideration of ESG issues in the academic literature. *Journal of Business Ethics*, 104(3), 389–402.
- Ethical Investment Association (EIA). (2005). *Sustainable responsible investment in Australia*. Sydney: EIA.
- Eurosif. (2010). *European SRI Study*, Revised Edition. Paris: Eurosif Studies.
- Financial Services Authority. (2006). *Private Equity: A discussion of risk and regulatory engagement*. http://www.fsa.gov.uk/pubs/discussion/dp06_06.pdf.
- Fisman, R., Heal, G., & Nair, V. B. (2007). Corporate Social Responsibility: Doing well by doing good? Working paper, Columbia University.
- Gaspar, J. M. (2009). The performance of French LBO firms: New data and new results. Working Paper, ESSEC, France. http://www.essec-private-equity.com/us/research/pdf/LBOperformance_abstract.pdf.
- Gompers, P., & Lerner, J. (2000). Money chasing deals? The impact of fund inflows on Private Equity valuation. *Journal of Financial Economics*, 55(2), 281–325.
- Grubbs, F. E. (1969). Procedures for detecting outlying observations in samples. *Technometrics*, 11(1969), 1–21.
- Hellmann, T. (2002). A theory of strategic venture investing. *Journal of Financial Economics*, 64(2), 285–314.
- Hellmann, T., Lindsey, L., & Puri, M. (2008). Building relationship early: Banks in venture capital. *Review of Financial Studies*, 21(2), 513–541.
- Holmstrom, B., & Tirole, J. (1997). Financial intermediation, loanable funds, and the real sector. *The Quarterly Journal of Economics*, 112(3), 663–691.
- Hull, C. E., & Rothenberg, S. (2008). Firm performance: The interactions of corporate social performance with innovation and industry differentiation. *Strategic Management Journal*, 29(7), 781–789.
- Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. *The American Economic Review*, 76(2), 323–329.
- Jensen, M. C. (1989). Eclipse of the public corporation. *Harvard Business Review*, 67, 60–70.
- Jensen, M. C. (1993). The modern industrial revolution, exit, and the failure of internal control systems. *Journal of Finance*, 48(3), 830–879.
- Jin, L., & Wang, F. (2002). Leveraged buyouts: Inception, evolution, and future trends. *Perspectives*, 3(6), 3–22.
- Johan, S., & Najar, D. (2010). The role of law, corruption and culture in investment fund manager fees. *Journal of Business Ethics*, 95(2), 147–172.
- Johnson, R. A., & Greening, D. W. (1999). The effects of corporate governance and institutional ownership types on corporate social performance. *Academy of Management Journal*, 42(5), 564–576.
- Kanter, R. M. (1999). From spare change to real change: The social sector as beta site for business innovation. *Harvard Business Review*, 77(3), 122–132.
- Kaplan, S., & Schoar, A. (2005). Private equity performance: Returns, persistence and capital flows. *Journal of Finance*, 60(4), 1791–1823.
- Kaplan, S. N., & Strömberg, P. (2009). Leveraged buyouts and Private Equity. *Journal of Economic Perspectives*, 23(1), 121–146.
- Kotler, P., & Lee, N. (2005). *Corporate Social Responsibility: Doing the most good for your company and your cause*. New York: Wiley.
- Kreander, N., Gray, R. H., Power, D. M., & Sinclair, C. D. (2005). Evaluating the performance of ethical and non-ethical funds: A matched pair analysis. *Journal of Business Finance and Accounting*, 32(7–8), 1465–1493.
- Kuckertz, A., & Wagner, M. (2009). The influence of sustainability orientation on entrepreneurial intentions—investigating the role of business experience. *Journal of Business Venturing*, 25(5), 524–539.
- Le Nadant, A. L., & Perdreau, F. (2006). Financial profile of leveraged buy-out targets: Some French evidence. *Review of Accounting and Finance*, 5(4), 370–392.
- Lerner, J., & Schoar, A. (2005). Does legal enforcement affect financial transactions? The contractual channel in Private Equity. *The Quarterly Journal of Economics*, 120(1), 223–246.
- Lewis, A., & Mackenzie, C. (2000). Support for investor activism among U.K. ethical investors. *Journal of Business Ethics*, 24(3), 215–222.
- Louche, C., & Lydenberg, S. (2006). Socially responsible investment: Differences between Europe and United States Working Paper, Vlerick Leuven Gent Management School.
- Lutz, S., Lyon, T. P., & Maxwell, J. W. (2000). Quality leadership when regulatory standards are forthcoming. *The Journal of Industrial Economics*, 48(3), 331–348.
- Lydenberg, S. D. (2002). Envisioning socially responsible investment: A model for 2006. *Journal of Corporate Citizenship*, 7, 57–77.

- Mahoney, L. S., & Thorne, L. (2006). An examination of the structure of executive compensation and Corporate Social Responsibility: A Canadian investigation. *Journal of Business Ethics*, 69(2), 149–162.
- Margolis, J. D., Elfenbein, H., & Walsh, A. (2009). Does it pay to be good... and does it matter? A meta-analysis and redirection of research on corporate social and financial performance. Working paper, Harvard University. http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1866371.
- McWilliams, A., Siegel, D. S., & Wright, P. M. (2006). Corporate Social Responsibility: Strategic implications. *Journal of Management Studies*, 43(1), 1–18.
- Moore, J. S., & Reichert, A. K. (1983). An analysis of the financial management techniques currently employed by large US corporations. *Journal of Business Finance and Accounting*, 10(4), 623–645.
- Morrell, K., & Clark, I. (2010). Private Equity and the public good. *Journal of Business Ethics*, 96(2), 249–263.
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions: When firms have information that investors do not have. *Journal of Financial Economics*, 13(2), 187–221.
- Neumann, W. L. (2003). *Social research methods*. Boston: Allyn and Bacon.
- Orlitzky, M., Schmidt, F. L., & Rynes, S. L. (2003). Corporate social and financial performance: A meta-analysis. *Organization Studies*, 24(3), 403–441.
- Porter, M. E., & Kramer, M. R. (2002). The competitive advantage of corporate philanthropy. *Harvard Business Review*, 80(12), 56–68.
- Porter, M. E., & Van der Linde, C. (1995). Toward a new conception of the environment–competitiveness relationship. *Journal of Economic Perspectives*, 9(4), 97–118.
- Portney, P. R. (2008). The (not so) new Corporate Social Responsibility: An empirical perspective. *Review of Environmental Economics and Policy*, 2(2), 261–275.
- Post, J. E., Preston, L. E., & Sachs, S. (2002). *Redefining the corporation: Stakeholder management and organizational wealth*. Stanford, CA: Stanford University Press.
- Reinhardt, F. L., Stavins, R. N., & Vietor, R. H. K. (2008). Corporate Social Responsibility through an economic lens. *Review of Environmental Economics and Policy*, 2(2), 219–239.
- Renneboog, L., Horst, J. T., & Zhang, C. (2008). Socially responsible investments: Institutional aspects, performance, and investor behavior. *Journal of Banking & Finance*, 32(9), 1723–1742.
- Scholtens, B. (2006). Finance as a driver of Corporate Social Responsibility. *Journal of Business Ethics*, 68(1), 19–33.
- Smith, M. (1996). Shareholder activism by institutional investors: Evidence from CalPERS. *Journal of Finance*, 51(1), 227–252.
- Social Investment Forum (SIF). (2003). *2003 report on socially responsible investing trends in the United States*. Washington, DC: Social Investment Forum.
- Social Investment Forum (SIF). (2010). *US SIF 2010 Annual Report*. Washington, DC: Social Investment Forum.
- Social Investment Organisation (SIO). (2005). *Canadian social investment review 2004*. Toronto: Social Investment Organisation.
- Sparkes, R. (1998, July). Through a glass darkly: Some thoughts on the ethics of investment. *Epworth Review*, 13–27.
- Sparkes, R. (2008). Socially responsible investment. In F. J. Fabozzi (Ed.), *The handbook of finance. Investment management and financial management* (Vol. 2). New York: Wiley.
- Sparkes, R., & Cowton, C. J. (2004). The maturing of socially responsible investment: A review of the developing link with corporate social responsibility. *Journal of Business Ethics*, 52(1), 45–57.
- Turban, D. B., & Greening, D. W. (1997). Corporate social performance and organizational attractiveness to prospective employees. *Academy of Management Journal*, 40(3), 658–672.
- Ueda, M. (2004). Banks versus venture capital: Project evaluation, screening, and expropriation. *Journal of Finance*, 59(2), 601–621.
- United Nations Principles for Responsible Investing (UNPRI). (2011). *Five years of PRI. Report on progress 2011*. UNEP-Finance Initiatives Publications.
- Vogel, D. (2005). *The market for virtue: The potential and limits of Corporate Social Responsibility*. Washington, DC: The Brookings Institution.
- Vyvyan, V., Ng, C., & Brimble, M. (2012). The influence of gender on the SRI decision making process: A study of attitudes and preferences using conjoint analysis. *Journal of Accounting and Finance*, 4, 17–34.
- Waddock, S. A., & Graves, S. B. (1994). Institutional owners and corporate social performance. *The Academy of Management Journal*, 37(4), 1034–1046.
- Williams, G. (2007). Some determinants of the socially responsible investment decision: A cross-country study. *Journal of Behavioral Finance*, 8(1), 43–57.
- Wright, M., & Robbie, K. (1996). Venture capitalists, unquoted equity investment appraisal and the role of accounting information. *Accounting and Business Research*, 26(2), 153–168.